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# On the interplay between clinical inertia and blood pressure measurement technology in hypertension management


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**Keywords:** *Hypertension management; chronic diseases; clinical decision making; clinical inertia; judgment bias*

## 1. Introduction

High blood pressure, aka hypertension (HTN), is a major public health issue globally (Chow and Gupta 2019). It is the most important risk factor for cardiovascular disease (CVD), especially heart attack and stroke (Yusuf et al. 2020), where CVD itself is the world's leading cause of death (Yusuf et al. 2020). Indeed, 77% of the first stroke events occur among patients with HTN (Go et al. 2014). The American College of Cardiology estimates that nearly half of adults in the US have HTN (Daniels 2019), and their total number worldwide is projected to reach 1.5 billion by 2025 (Williams et al. 2018). In the US, the estimated annual cost associated with HTN is $55.9 billion (Benjamin et al. 2019).

The good news is that HTN can be controlled through careful disease management, with substantial evidence supporting the clinical benefits of controlling HTN (Yusuf et al. 2020). For example, a 10-unit reduction in blood pressure (BP) lowers CVD risks by 20% and all-cause mortality by 13% (Wolf et al. 2018). The unfortunate news is that fewer than half of patients with HTN have the disease under control due to various reasons. First, HTN remains asymptomatic until the person faces severe health issues, such as CVD or kidney disease, earning the nickname "silent killer" in the medical community. According to the National Health Service, 5.6 million people in the UK are living with HTN without being aware of it (NHS 2017).

Second, BP fluctuates both in the short- and long-term (Mehlum et al. 2018), making consistent and reliable measurements quite difficult. Third, the conventional BP measurements are often noisy. These can introduce uncertainty in BP readings, as they may also vary due to different other factors such as stress, white coat effect, or temporary physiological changes (Rubin 2023). Thus, clinicians do not have high confidence in BP measurement, leading to subjectivity in clinical decision-making, reduced compliance with medical guidelines, and significant variation in clinical practices (Lebeau et al. 2018, Zargoush et al. 2018).

A growing body of literature, including the European HTN management guideline, recognizes *clinical inertia* as a major contributor to poor HTN control (Williams et al. 2018). This phenomenon refers to healthcare providers' failure or reluctance to recognize the elevated BP despite evidence to the contrary. It leads to diagnostic and treatment inertia, implying a lack of timely action in diagnosing the patient as hypertensive and intensifying medications. Many consider clinical inertia a "human characteristic" issue related to the physician's judgment bias (American Medical Association 2020, Lebeau et al. 2014), which is quite pervasive. Studies suggest that physicians' judgment is influenced by uncertainty in determining a patient's true BP, contributing to clinical inertia and judgment bias, which lead to suboptimal clinical decision-making (Ali et al. 2021). The reason is that clinicians often delay necessary changes in treatment while waiting for more definitive BP estimations (Muldoon et al. 2018). Augustin et al. (2021) highlight BP measurement errors and inherent BP variability as the root causes of clinical inertia, as clinicians may hesitate to act on readings they perceive as unreliable. Indeed, clinical inertia manifests in the physicians' "*wait until next visit* " approach to avoid treatment intensification, which has become a serious barrier to optimal HTN control (Ogedegbe 2008). Therefore, the accuracy and reliability of BP measurement are crucial for the proper diagnosis and treatment of HTN. A mere 5-unit error in BP measurement leads to HTN misclassification in millions of individuals worldwide, which results in inappropriate alterations to clinical decisions in 20% to 45% of cases (Padwal et al. 2019). In a study conducted with 7,253 hypertensive patients, medications were not intensified in 87% of the visits where an elevated BP was observed (Faria et al. 2009). Recent studies have reported similar statistics regarding poor medication prescriptions for HTN control due to clinical inertia (Koopman et al. 2020, Niriayo et al. 2024). Faria et al. (2009) report that only 1/3 of the surveyed physicians believed BP measurements were accurate. The Joint National Committee (JNC) report on the prevention, detection, evaluation, and treatment of HTN highlights the importance of addressing clinical inertia in HTN management (Gil-Guillen et al. 2010). By enhancing clinicians' confidence in their patients' BP, accurate measurement can effectively reduce clinical inertia and improve long-term HTN control.

Using traditional devices to measure BP multiple times may provide some benefits in terms of obtaining average values. However, it is important to note that relying on inaccurate devices, even if used repeatedly, may still lead to unreliable readings, resulting in misinterpretation of the patient's true BP status and physicians' judgment bias (such as clinical inertia) in their decision-making. Due to this dual impact, accurately assessing the effect of physicians' judgment bias on their clinical decisions regarding HTN requires considering technology-related variables, such as measurement noise. The interplay between human and technology factors means neither can be fully understood in isolation, especially concerning patient outcomes.

One potential solution to reduce BP measurement uncertainty and its ramifications, such as clinical inertia, is to improve measurement accuracy through modern technologies, such as automated office BP (AOBP) measurement. AOBP is a non-invasive technology that has been clinically validated for its accuracy and reliability (Roerecke et al. 2019). Several guidelines recommend using these technologies in clinical settings instead of traditional cuffs around the arm (Whelton et al. 2018, Williams et al. 2018). Since 2016, the Canadian HTN guideline has recommended AOBP as the preferred technology for BP measurement (Myers 2016). In the Systolic Blood Pressure Intervention Trial, AOBP was used to ensure measurement accuracy (SPRINT Research Group 2015). The MAP (Measure Accurately, Act Rapidly, and Partner with Patients) protocol developed in 2018 also highlights AOBP as a preferred method for improving BP measurement accuracy. This protocol demonstrated that AOBP reduces measurement noise, enabling physicians to make more accurate and timely decisions, which is crucial for mitigating clinical inertia (Egan et al. 2018).

Despite promising evidence, the adoption of these technologies has been slow, possibly because they are five times more expensive than traditional approaches, require more time, and need a private examination room. Such barriers lead clinicians to question whether the additional benefits of new technologies justify their costs (Roerecke et al. 2019). However, such perception often overlooks the positive impact of these technologies on the quality of clinical decisions regarding patient outcomes, particularly in relation to how they affect the physician's judgment bias.

The complex interaction between technology and physician judgment highlights the importance of innovative research in HTN management from a clinical decision-making standpoint. Our study focuses on understanding how physician judgment, such as clinical inertia, influences decision-making process and its quality (regarding patient outcomes) while controlling for technological impacts. This requires integrating rich and relevant clinical data with careful modeling that accurately reflects the decision-making process, considering both human and technology factors.

Shedding light on this knowledge gap is central to the objectives of the present study. Our first aim is to assess the impact of the human factor, specifically clinical inertia as a physician judgment bias, on decision-making policies in HTN management while considering the effect of technological factors such as measurement noise. Our study addresses this issue through an analytical framework with theoretical results that are calibrated with real clinical data, providing new insights into prescription decision-making in the context of clinical inertia. The second objective delves into the empirical evaluation of the joint impact of human and technology factors on decision qualities in terms of patient outcomes. We achieve this by examining the Decision Quality Decrement ($DQD$) caused by the human factor (HF) and technology factor (TF). We assess decision quality from the perspective of the patient's quality of life.

The conceptual framework for achieving our study's objectives is depicted in Fig 1. Based on the preceding discussion, $DQD$ is decomposed into two components: one component, $DQD_{HF}$, captures human factor, and the other component, $DQD_{TF}$, captures technology (see (5) in Fig 1). We estimate $DQD_{TF}$ by comparing the maximum payoffs (under optimal decision-making) obtained from unbiased learning (see (3)-(4) in Fig 1) and the perfect BP information (see 1 in Fig 1). To quantify $DQD_{HF}$, we compare the maximum payoffs obtained from unbiased and biased learning, where the latter incorporates physician's judgment bias (see (2) in Fig 1). This study focuses on medication (i.e., antihypertensive) treatment due to its critical role when lifestyle changes fail. This study addresses the following research questions:

**RQ1: Impact of Clinical Inertia on Decision-Making Policy:** *How prescription decisions about antihypertensive medications are made from the perspective of clinical inertia (physician judgment bias)?*

**RQ2: Interactive Impact of Human and Technology Factors on Decision Quality:** *What is the combined impact of human factors and technological factors on the decision-making quality in terms of patient quality of life?*

**RQ3: Moderating Factors in Decision Quality:** *How is the human-technology interactive effect moderated by other considerations, such as the sources of BP uncertainty, patient characteristics, and decision flexibility?*

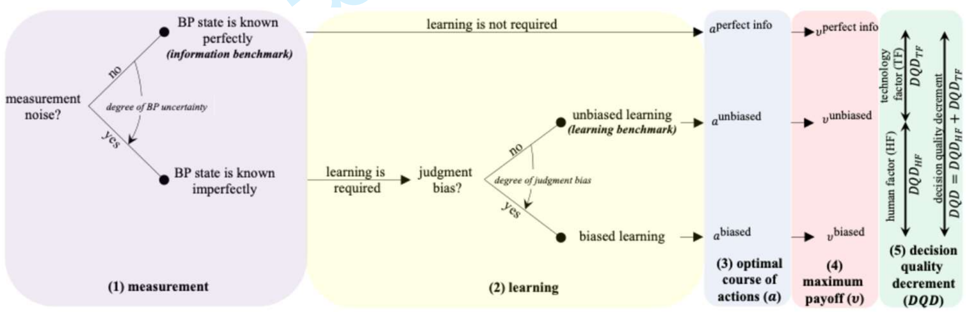


**Fig 1** The conceptual framework of our study

We present an analytical framework that simultaneously incorporates the essential features of the clinical HTN control process. This involves two main analytical modules that operate jointly. The first is the *learning module*, which formulates how clinical judgments regarding the patient's BP are formed (§3.2). We introduce surprise-induced learning (SIL) to simulate physician's learning (i.e., belief updating) under judgment bias as a human factor attribute. The judgment bias can manifest in either direction, i.e., over-reaction or under-reaction to the BP measurements (Mukherjee and Sinha 2018). Over-reaction refers to assigning disproportionate weight to new evidence, leading to premature adjustments, while under-reaction involves insufficient responsiveness to new evidence, potentially delaying necessary interventions. Therefore, the under-reaction bias aligns with the concept of clinical inertia, where clinicians are reluctant to add new medications or confirm HTN by under-valuing the current BP measurement. This aligns with experimental studies on human learning psychology, consistently showing that under-reaction is a more common cognitive bias than over-reaction during sequential learning (Benjamin 2019). Moreover, a study investigating judgment bias in product recall decisions has reported similar findings (Mukherjee and Sinha 2018).

The second element of our analytical framework, the *optimization module*, involves a Markov decision process (MDP) model that formulates how prescription decisions are optimized under BP uncertainty and judgment bias (§4). In this model, the optimal course of action maximizes CVD risk reductions while minimizing medication side effects, manifesting in the maximized patient's overall quality of life. The optimal policies take into account physician judgment and are tailored to patient characteristics associated with CVD risk as well as short- and long-term BP variability, which are critical for prescription decisions (Hsu et al. 2016, Narita et al. 2023). Clinical guidelines highly recommend prescribing antihypertensive medications based on patient's CVD risk instead of targeting certain BP levels (Scott et al. 2019, Williams et al. 2018).

Our research is grounded in a robust analytical framework that reflects current medical practice, thus improving the face validity of our models and increasing the potential for acceptance and implementation of the insights gained from our results. In addition to the literature, to ensure that our model accurately reflects physicians' cognitive learning in HTN management, we surveyed physicians across various specialties related to HTN management. Each question in the survey was designed to verify a specific component of our learning model, providing us with valuable insights into the real-world applicability of our modeling constructs. This approach not only strengthens the credibility of our research but also bridges the gap between theory and practice, ensuring that our findings are both relevant and actionable in a clinical setting. Moreover, we calibrate our models with the primary data collected under a unique clinical setting in the Montreal General Hospital's Vascular Health Unit, where a rigorous protocol is applied to use modern technologies (AOBP) to measure BP, which we consider a reasonable proxy for accurate BP measurements.

**Main Results and Contributions**: Our study establishes that optimal prescription under clinical inertia follows structured threshold policies. These policies formalize how physicians can balance the trade-off between exploration (delaying treatment to gather more information) and exploitation (acting on available information). Our results reveal that while some delays in treatment can be optimal (allowing clinicians to make more informed decisions), prolonged delays often become suboptimal, leading to adverse patient outcomes. Such understanding provides first-hand evidence to address the critical gap in HTN management caused by clinical inertia, offering actionable strategies to refine treatment protocols. Regarding decision quality, our study highlights the complexity of the main and interactive effects of HF and TF on DQD. These effects are significantly influenced by other critical factors, including patient's BP variability, CVD risk profile, and physician decision flexibility. Notably, the interaction between HF and TF is particularly significant, emphasizing the need to address both simultaneously when aiming to mitigate the detrimental impact of clinical inertia on decision quality. These insights provide a comprehensive understanding of how decision-making and decision quality are shaped by multiple interacting factors, offering practical recommendations for overcoming clinical inertia and optimizing treatment protocols in HTN management.

This research makes several contributions to the operations research and management science (OR/MS) domain by explicitly investigating the interplay between physician judgment and technology in chronic disease management. First, we develop analytical models that provide theoretical insights into optimal decision-making by incorporating physician judgment biases and measurement noise. The structured threshold policies derived from these models offer actionable strategies to improve HTN treatment decisions. Second, by examining the dependencies of DQD on factors such as BP variability, patient sex, and decision flexibility, our study highlights how contextual factors amplify or mitigate the effects of HF and TF. Third, introducing relevant concepts, such as clinical inertia, modeled through the notions of surprise and SIL, represents a key contribution of this study. These concepts integrate cognitive and behavioral dimensions of clinical decision-making into our framework, enhancing its clinical relevance and practical applicability. Fourth, by addressing clinical inertia as a critical barrier to effective HTN management, this research deepens our understanding of its pervasive effects on treatment outcomes. We provide actionable insights into balancing the exploration-exploitation trade-off, thereby optimizing decision-making and countering the "wait until next visit" approach highlighted in prior literature.

## 2. Clinical Context and Relevant Literature

### 2.1. Hypertension Measurement and Treatment

BP is measured in millimeters of mercury unit, denoted by mmHg, and comprises systolic (pressure during the heartbeat) and diastolic BP (pressure between heartbeats). There is strong evidence that systolic BP is a better predictor of CVD risk and a preferred indicator of BP (Roerecke et al. 2019). In the rest of this paper, we will refer to BP as systolic BP. HTN is diagnosed when there is a consistent BP elevation, although the criteria for diagnosis vary significantly among existing guidelines (Karmali et al. 2018, Zargoush et al. 2018). Elevated BP over a prolonged period can damage vital organs, such as the heart, brain, and kidneys. This study focuses on medication therapy because of its crucial role in managing HTN when lifestyle modifications alone are insufficient. Evidence shows that patients who struggle to achieve adequate BP control through lifestyle changes, like maintaining a healthy diet, engaging in regular physical activity, and quitting smoking, often need medication intervention to reduce the serious risks associated with HTN. Notably, many hypertensive patients fail to make necessary lifestyle changes, emphasizing the importance and prevalence of medication therapy in HTN management (Charchar et al. 2024, Whelton et al. 2018).

Unfortunately, approximately two-thirds of hypertensive patients do not achieve the desired BP levels with a single drug. Most of these patients are required to take multiple medications (aka combination therapy) daily for the rest of their lives (Crawford 2019, Smith et al. 2020). In this study, we consider the following five generic classes of antihypertensives that vary in their effectiveness for BP reduction and their side effects: angiotensin-converting enzyme inhibitors (ACEIs), angiotensin II receptor blockers (ARBs), beta-blockers (BBs), calcium channel blockers (CCBs), and diuretics (DUs), with thiazide being the most common type.

### 2.2. Heuristic Thinking and Cognitive Bias in Clinical Decision-Making

In decision-making, individuals often face cognitive limitations and uncertainty, which require them to make judgments and choices based on existing information, which is often incomplete or inaccurate. To simplify this process, they rely on cognitive shortcuts known as heuristics (unconscious mental shortcuts) to save time and effort for decision-making (Raue and Scholl 2018). In a seminal work, Kahneman (2011) suggests that human beings typically do not engage in complex mental operations to integrate and process various pieces of information for decision-making. Instead, they frequently depend on heuristics. There is a downside to using heuristics, as they can sometimes lead to systematic cognitive errors known as biases. These biases can lead to errors in judgment and decision-making, particularly when people do not actively seek out additional information or consider all relevant factors. They can negatively affect probability estimation and information synthesis, which are central to clinical decision-making (Blumenthal-Barby and Krieger 2015).

Over 100 distinct biases have been reported in the context of clinical decision-making (Webster et al. 2021). Among these, three biases are of particular interest to our study. They include (a) "anchoring and adjustment," a bias where the emphasis is placed on the initial piece of information received during decision-making, (b) the "overconfidence bias," which reflects individuals' tendency to over-reaction to their knowledge and underweight their ignorance, and (c) "commission bias," a tendency to seek out and favor information that confirms pre-existing beliefs. Such cognitive processes influence decisions in routine clinical care, with potential implications for patients' treatments and outcomes. Research on cognitive biases and heuristics in medical care and their impact on medical decision-making is becoming increasingly important.

Blumenthal-Barby and Krieger (2015) offer a foundational understanding of the prevalence of cognitive biases among clinicians, raising concerns about the relevance of the findings to actual decision-making due to the reliance on hypothetical vignettes in most studies. This concern is echoed in several other studies, underscoring the need for more real-world research. For instance, Delgadillo et al. (2017) and O'Sullivan and Schofield (2018) highlight the complexity of clinical decision-making processes, suggesting that they are influenced by many factors, including beliefs, attitudes, subjective norms, and perceptions of self-efficacy. This complexity is further complicated by cognitive biases, which can result in diagnostic errors and inconsistencies in treatment planning. Featherston et al. (2020) and Saposnik et al. (2016) extend this discussion to health professionals and physicians, suggesting that cognitive biases and personality traits can lead to diagnostic inaccuracies and medical errors. These findings underscore the need for interventions to mitigate the impact of these biases in clinical practice. Kc (2020) provides a real-world example of how heuristic thinking and cognitive bias can lead to increased length of stay in acute care hospitals, highlighting the practical implications of these biases. Similarly, Erel et al. (2023) discuss the influence of cognitive biases on healthcare workers' decision-making, emphasizing the potential impact on patient outcomes. Esposito (2023) and Michael et al. (2023) suggest that cognitive biases and heuristics may negatively impact diagnosis and treatment planning, underscoring the need for strategies to combat biases. Gorini and Pravettoni (2011) provide a comprehensive assessment of the role of cognitive biases and heuristics in medical decision-making, arguing for the development of tools to assist in the decision-making process. This is echoed by Richburg et al. (2023), who highlight the potential of AI in mitigating the impact of cognitive biases in surgical decision-making. Finally, Staats et al. (2018) discuss the significant impact of cognitive biases on clinical decision-making in ICUs, emphasizing the need for strategies to mitigate these biases.

The comprehensive literature reviewed here underscores the pervasive influence of cognitive biases and heuristics on medical decision-making across various healthcare settings and disciplines. While some studies have explored the intersection of physician decision-making biases and technology in healthcare, this body of research reveals a notable gap in understanding the specific interplay between physician judgment and technological factors, such as measurement noise, in the context of HTN management (Dai et al. 2022, Staats et al. 2018). This gap, to be addressed by the proposed study, is especially evident when considering the impact of these factors on how clinical decisions are made and their quality regarding patient well-being.

**2.3. Decision-Making for HTN Management**

Schell et al. (2016) studied the personalization of HTN treatment plans by integrating various CVD-related health states into an MDP framework. Using similar state variables, Schell et al. (2019) optimized co-insurance payments to improve hypertensive patients' adherence to their treatment plans. These two studies did not provide BP-dependent optimal prescription policies. Lee et al. (2018) investigated optimizing the intervention programs for change in lifestyle (such as diet, exercise, smoking cessation, and drinking moderation) to prevent HTN. Bonifonte et al. (2022) adopted stochastic optimization for the antihypertensive prescription at the population level. Their study minimizes the BP risk of mortality quantified by the so-called J-curve function. In a general framework, Keyvanshokooh et al. (2019) developed a joint learning-optimization algorithm for managing chronic diseases. They applied their algorithm to optimize the third-line medication treatment for hypertensive patients with type II diabetes.

The studies most related to our study are two foundational studies, notably those by Zargoush et al. (2018) and Garcia et al. (2023), which utilized MDP to optimize antihypertensive prescriptions. They primarily focused on establishing optimal threshold policies for prescriptions, assuming noise-free BP measurements. However, our study significantly diverges in both methodology and application by integrating several critical factors that have been previously overlooked in the literature. We consider the multifaceted impacts of technology, such as measurement noise, as well as human elements, including physician learning and judgment bias. Moreover, our research uniquely addresses variability in BP, both in the short- and long-term, which are crucial aspects of managing HTN but have largely been absent from existing studies. Our approach is more comprehensive, examining not only the profoundly distinct formulation of optimal clinical decisions and their structural properties in a more realistic setting but also evaluating the influence of these factors and their interactions on decision policy and quality. This holistic perspective enables our study to more accurately represent actual medical practice, offering insights that are distinct, unique, and better aligned with real-world clinical environments. This approach not only fills a gap in the current body of knowledge but also provides a deeper understanding of HTN management.

**2.4. Bayesian Belief Updating in Clinical Decision-Making**

In clinical decision-making, Bayesian belief updating involves adjusting the likelihood of a patient having a certain condition based on new test results or symptoms (Austin 2019). Achtziger et al. (2014) highlight that humans, including physicians, often deviate from Bayesian belief updating as the normative model by either overweighing prior information (conservatism) or new information (representativeness heuristic). Rottman (2017) found physicians often overestimate the post-test probability of a disease, especially when the pre-test probability and the likelihood ratio of the test are low. This overestimation seems consistent across various medical tests and diseases, suggesting that physicians update their beliefs in a Bayesian manner when given diagnostic test results. However, they often overestimate the post-test probability of disease while underutilizing the prior odds and likelihood ratio. Brush et al. (2019) examine whether novice clinicians could be taught to make more accurate Bayesian revisions of diagnostic probabilities. They found that students benefited from theoretical instruction on Bayesian concepts. All participants' probability estimates were, on average, close to the Bayesian calculation. Austin (2019) discussed using Bayesian reasoning in physician interpretation of test results. The article suggests that physicians reason in a way that is roughly consistent with Bayes' law, intuitively consider base rates, and recognize that the positive predictive value in a diagnostic test is higher than in a screening test. However, the general practitioners in the study were not adequately calibrated as Bayesians; they perceived screening tests more like diagnostic tests than they truly are. This indicates that physicians may be relying on "fast and frugal" heuristics, which are not sufficiently calculative.

These studies confirm that although physicians utilize Bayesian belief updating in their decision-making, they often deviate from the normative models. Our study will contribute to this literature by modeling physician learning with a modified Bayesian belief updating approach, which offers a deeper insight into physicians' decision-making. To this end, our research represents a groundbreaking effort to introduce the concept of surprise in dynamic Bayesian learning, capturing how unexpected clinical developments, such as sudden changes in a patient's BP, influence physicians' judgment and decisions. This innovative approach aligns with the dynamic and often unpredictable nature of HTN, where patient's BP can vary considerably from expectations. Our learning model is designed for a non-stationary element (here, patient's true BP), recognizing the evolving nature of a patient's health status, hence enabling more adaptive treatment strategies over time. The presence and significance of surprise in clinical practice were validated through clinical literature and our comprehensive survey (detailed in §3.4, 5.2), confirming its relevance to real-world HTN management. The survey results suggest that incorporating surprise into our model enhances theoretical understanding and practical applications, especially for managing chronic conditions like HTN.

## 3. BP Measurement and Learning

In this section, we describe the stochastic environment in which BP evolves and is observed by physicians. Next, we propose a learning mechanism that simulates how physicians learn, i.e., adjust their judgment (belief) about a patient's BP, which we refer to as surprise-induced learning (SIL). The last part briefly presents

Kalman Filter (KF) as the learning benchmark. These components provide the necessary infrastructure for the learning module within our analytical framework. The list of notations is in Appendix B.

### 3.1. BP Progression

A key challenge in managing HTN is the inability to directly observe the true underlying BP. However, information about the underlying BP can be obtained through sequential BP observations at the clinic. To formulate BP progression in the presence of noisy measurements under the short- and long-term BP variability, we adopt a Markov chain representation of BP with a hidden layer (Fig 2). The hidden layer, denoted by $\theta_t$, represents the patient's true BP mean in period $t$, which evolves according to the stochastic recurrence relation $\theta_{t+1} = \theta_t + e_\theta$. In this recurrence equation, $e_\theta \sim \mathcal{N}(\mu'_\theta, \sigma_\theta^2)$ captures long-term change in the BP regimen, which is affected by two main factors. The first factor, denoted by $\mu'_\theta$, can be attributed to known causes, such as aging (Yano 2017) and medications' impact on BP (Law et al. 2009). The second factor captures random factors, denoted by $\sigma_\theta^2$, which cannot be attributed to any known causes and corresponds to the long-term BP variability (LBPV), aka visit-to-visit BP variability in medicine (Stevens et al. 2016).

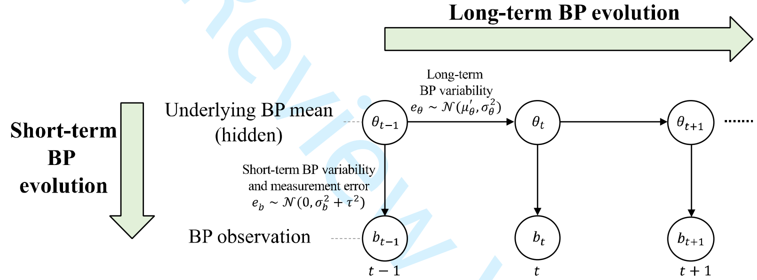


**Fig 2** The stochastic model of BP evolution

Given the patient's underlying BP regimen $\theta_t$, we consider the patient's observed BP in $t$, denoted by $b_t$, alongside two sources of variabilities: short-term BP variability (SBPV) and measurement error. The former, which is a clinically established concept (Stevens et al. 2016), captures BP variability around the underlying BP regimen $\theta_t$ in a short time, e.g., hourly, and the latter results from measurement noise. The uncertainty caused by these two sources can be represented by a normally distributed random variable with a mean of zero and variance $\sigma_b^2 + \tau^2$. Thus, we can formulate the BP process as the following linear dynamic system:

$$\begin{cases} \textbf{Observation Equation:} & b_t = \theta_t + e_b; \quad e_b \sim \mathcal{N}(0, \phi^2) \\ \textbf{State Equation:} & \theta_t = \theta_{t-1} + e_\theta; \quad e_\theta \sim \mathcal{N}\left(\mu'_\theta, \sigma_\theta^2\right) \end{cases} \tag{1}$$

$\phi^2 = \sigma_b^2 + \tau^2$ represents total variance. This formulation is suitable for modeling chronic diseases where the underlying hidden state includes process noise and the medical examinations exhibit measurement noise (Helm et al. 2015). Our assumption regarding the stationarity of both $\sigma_\theta^2$ and $\sigma_b^2$ is backed by the medical literature (Goh et al. 2017). However, this does not imply the underlying BP process is stationary. Indeed, we consider it to be non-stationary. In practice, physicians observe $b_t$ and use it to form their beliefs, serving as their best estimates of the patient's underlying BP regimen $\theta_t$ (learning module). Finally, based on their beliefs about $\theta_t$, physicians decide *if*, *when*, and *which* medication should be prescribed (decision-making module).

### 3.2. Physician's Learning under Judgment Bias

Informed by the extensive literature confirming the use of Bayesian belief updating in medical decision-making (Achtziger et al. 2014, Austin 2019, Brush et al. 2019, Rottman 2017), we now describe how physicians form and update their beliefs about the underlying BP regimen based on the patient's BP history and the BP readings at each visit. To formulate this process, we use a modified version of the standard Bayesian belief updating mechanism. For now, we focus on the stochastic components of BP evolution and will integrate everything when presenting the optimization models. Let $\pi_t^B(\theta_t)$ denote the distribution of the beliefs about the underlying BP regiment in $t$, obtained through the Bayesian updating mechanism:

$$\pi_t^B(\theta_t) \equiv \pi_t^B(\theta) = \mathcal{N}\left(\mu_t^B, \sigma_{t,B}^2\right) \tag{2}$$

Using Bayes' theorem and the self-conjugacy property of normal distribution, the updating equations for the belief distribution parameters can be presented as follows:

$$\begin{cases} \mu_t^B = \rho_t b_t + (1-\rho_t)\mu_{t-1}^B \\ \sigma_{t,B}^2 = (1-\rho_t)\sigma_{t-1,B}^2 \end{cases}; \; where\ \rho_t = \frac{\sigma_{t-1,B}^2}{\sigma_{t-1,B}^2 + \phi^2} \tag{3}$$

where, $\mu_t^B$ and $\sigma_{t,B}^2$ are the mean and variance of the belief distribution. Given the new BP observation $b_t$, $\mu_t^B$ is the best estimate of the underlying BP in $t$ and $\sigma_{t,B}^2$ represents the uncertainty associated with the estimate. Moreover, parameter $\rho_t \in [0,1]$ is called the *learning rate* and represents the relative weight assigned to the new BP observation compared to the patient's BP history. As such, the updated belief regarding the underlying BP mean is a weighted average of the prior BP history $\mu_{t-1}^B$ and the new BP observation $b_t$.

The above learning procedure in the standard Bayesian assumes that the underlying process is stationary. It suggests that as the number of patient visits increases, the updated belief converges to a constant value and the uncertainty surrounding that belief decreases, regardless of the changes in the underlying BP regimen. The stationarity assumption does not hold true in real practice of HTN management because BP regimen changes over time. We propose a learning mechanism that responds to BP readings, which may indicate a change in the BP regimen. The proposed mechanism, which is also strongly supported by clinicians based on our survey (§3.4 and §5.2), incorporates subjective judgment (bias) into the learning process through the under-reaction or over-reaction to the new information. To this end, we introduce the notion of *surprise*, denoted by $s_t$, as a binary variable that equals one if the difference between the prior belief and the new BP measurement is beyond a certain threshold level Δ; otherwise, it equals zero. Therefore, we formulate surprise as follows:

$$s_t = \begin{cases} 1, & \text{if } |\mu_{t-1} - b_t| \geq \Delta \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

The threshold $\Delta \in [0, \infty)$ determines the significance of the difference between prior beliefs about the underlying BP and the new evidence $b_t$. If the difference is greater than or equal to Δ, the new BP reading is considered a surprise, signaling a change in the BP regimen.

The notion of surprise has played a pivotal role in the context of cognitive learning and decision-making in understanding how individuals adapt to sudden changes and disruptions in their decision-making. For example, Acevedo et al. (2022) provide a framework to analyze the impact of surprise on consumer behavior and decision-making. They show that disruption caused by the pandemic made consumers reevaluate their consumption choices, as evidenced by the shifts in their market baskets. This reevaluation mirrors the cognitive learning process where surprises necessitate a reassessment of previously held beliefs and preferences. This aligns with existing literature on cognitive learning, emphasizing the role of unexpected events (surprise) in triggering cognitive adjustments in learning. The concept of surprise in cognitive psychology, often linked to unexpected stimuli, prompts individuals to reassess and update their knowledge and behaviors (Schneider and Shiffrin 1977). Foster and Keane (2019) found that unexpected results can improve memory retention by increasing attention. This also aligns with the Ash (2009)'s examination of surprise in hindsight bias, demonstrating that unexpected outcomes significantly affect retrospective judgment and memory recall. Standing (2008) applied cognitive continuum theory to clinical judgment in nursing, illustrating how different types of judgment tasks, influenced by surprise, affect decision-making. de Gee et al. (2020) showed that physiological markers such as pupil dilation and event-related potentials reflect subjective surprise (i.e., the mismatch between one's expectation and the observed outcome) in decision-making, providing a measurable indicator of cognitive responses to unexpected outcomes.

Similar concepts of surprise have been introduced in machine learning literature. One is Shannon entropy (Shannon 2001), defining surprise as the likelihood of evidence, hence, observing a rare event indicates surprise (Faraji et al. 2017). Another notion is Bayesian surprise, defined as the difference (measured in terms of likelihood using the Kullback-Leibler divergence) between the prior belief $\pi_{t-1}(\theta)$ and the posterior belief $\pi_t(\theta)$ (Itti and Baldi 2009). While both notions center on the likelihood of events in their definitions of surprise and describe it objectively, we define surprise as a subjective concept rooted in the physician's prospect over what they *expect* to observe. An expectation-based definition of surprise has also been introduced in the context of human learning (Faraji et al. 2017), but it leads to a more complex formulation of surprise on a continuous scale. Our definition is more straightforward and intuitive because it only compares prior beliefs with new observations, resulting in a binary definition of surprise (Eq. 4). This definition better serves our analytical purposes and makes it more appealing for applications in clinical decision-making.

Reinforcement learning models have also integrated surprise to enhance decision-making. Loftus et al. (2020) utilized reinforcement learning for surgical decision-making. They discuss decision-making challenges faced under uncertainty and time constraints in surgery, highlighting how cognitive and judgment errors can occur. This ties into the notion of surprise, as unexpected outcomes disrupt expected decisions, leading to cognitive reassessment. They used reinforcement learning as a tool to help manage unexpected outcomes by learning from past experiences to optimize future decisions. Incorporating reinforcement learning implies dealing with prediction errors, which are essentially surprises that occur when there is a discrepancy between expected and actual outcomes. Like our proposed learning, this model incorporates the ability for surprise to adjust learning rates. When an expected outcome differs from the actual outcome, the prediction error (i.e., surprise) signals the system to adjust decisions. This mechanism is crucial for adapting to unexpected scenarios, thereby improving the accuracy and reliability of surgical decision-making over time. Sumiya and Katahira (2020) proposed a surprise-sensitive utility model, emphasizing the dynamic adjustment of learning rates in response to surprises as their mechanism for adapting to unexpected events.

In our proposed learning strategy, when a surprise occurs, a *shock* is imposed on the physician's belief system, readjusting $\sigma_t^2$ and $\rho_t$ to their initial (i.e., max) values. This readjusts the relative weight of the new evidence and the uncertainty surrounding the new belief such that i) maximum attention is given to the evidence that triggered the surprise and ii) the new belief has highest uncertainty. We refer to this learning strategy as *surprise-induced learning* (SIL). This learning mechanism formulates how clinicians reconcile new information with their prior expectations, highlighting the complex dynamic of clinical inertia. Similarly, Goodwin et al. (2023) explore Bayesian belief updating in perceptual decision-making, proposing that the brain constructs an internal model to integrate prior expectations with incoming observations, weighted by their respective uncertainties, to estimate the states of an environment. They argue that accurately integrating these information sources is crucial for facilitating effective perceptual decision-making.

Let $\pi_t^{SIL}(\theta_t)$ denote the belief distribution regarding the underlying BP regimen attained through SIL:

$$\pi_t^{SIL}(\theta_t) = \mathcal{N}\left(\mu_t^{SIL}, \sigma_{t,SIL}^2\right) \tag{5}$$

Incorporating surprise, we modify the updating mechanism parameters as follows:

$$\begin{cases} \mu_t^{SIL} = \rho_t b_t + (1-\rho_t)\mu_{t-1}^{SIL}; & \sigma_{t,SIL}^2 = (1-\rho_t)\sigma_{t-1,SIL}^2 & \text{if } \mathcal{s}_t = 0 \\ \mu_t^{SIL} = \rho_1 b_t + (1-\rho_1)\mu_{t-1}^{SIL}; & \sigma_{t,SIL}^2 = (1-\rho_1)\sigma_{0,SIL}^2 & \text{if } \mathcal{s}_t = 1 \end{cases} \tag{6}$$

where $\rho_t$ is quantified by Eq. (3). Under SIL, if a surprise occurs, the previous BP history will not be entirely discarded. Instead, it will be given a minimum (non-zero) weight in response to the observed signal concerning the BP regimen change, triggering maximal attention to the signal. This mechanism also enables a return to the previous regimen, although at a slower rate, if the observation was merely a random event.

In our study, the threshold parameter is crucial as it characterizes physicians' learning behavior and responsiveness to BP observations. Physicians with low $\Delta$ experience more frequent surprises because they have a lower bar for change. Such physicians view even a minor change in BP as unexpected, assigning larger weights to the new observations than their prior beliefs. Physicians with higher $\Delta$ become surprised less frequently because they have a higher bar to consider a BP change significant, thus assigning larger weights to their beliefs. The parameter $\Delta$ can also be interpreted as the physician's degree of *commitment to belief* (Itti and Baldi 2009), where a low (resp. high) commitment to belief represents an over-reaction (resp. under-reaction) of the significance of the new BP measurement. This is reflected in the physician's bias in taking prescriptive actions. For example, higher $\Delta$ may result in a delay in medication prescriptions, whereas lower $\Delta$ may result in premature medication prescriptions. Therefore, the overcommitment to belief, which leads to underestimating new evidence, serves as a reasonable proxy for physician inertia.

### 3.3. A Learning Benchmark

As a benchmark for the learning process, we use KF, which has proven to be an optimal learner for the system in Eq. (1). KF is considered unbiased because it corrects predictions based on new observations, ensuring no systematic deviation from the true state, assuming zero-mean Gaussian noise in the process and measurements. This means that the expected value of the estimation error remains zero over time, allowing the estimated state to closely match the actual state under reasonable assumptions (Helm et al. 2015, van Lint and Djukic 2012). This unbiasedness is mathematically guaranteed by the prediction-correction mechanism and is empirically validated through minimal prediction errors, as demonstrated by Helm et al. (2015) and van Lint and Djukic (2012). From a non-technical perspective, the KF is unbiased because it relies solely on mathematically-defined parameters, avoiding the inclusion of subjective parameters, ensuring objectivity in the estimation process and independence from human-induced bias. KF has been extensively used in clinical decision-making (Helm et al. 2015, Kazemian et al. 2019).

Let $\pi_t^{KF}(\theta_t)$ denote the belief distribution regarding the underlying BP regimen, such that

$$\pi_t^{KF}(\theta_t) = \mathcal{N}\left(\mu_t^{KF}, \sigma_{t,KF}^2\right) \tag{7}$$

where $\mu_t^{KF}$ and $\sigma_{t,KF}^2$ are the best estimates of the underlying BP in $t$ and the uncertainty associated with this estimate. After observing a new BP measurement $b_t$, the belief distribution parameters are updated according to the KF algorithm as follows (Chui and Chen 2017):

$$\begin{cases} \mu_t^{KF} = K_t b_t + (1 - K_t)\mu_{t-1}^{KF} \\ \sigma_{t,KF}^2 = (1 - K_t)\left(\sigma_{t-1,KF}^2 + \sigma_\theta^2\right) \end{cases}; where\ K_t = \frac{\sigma_{t-1,KF}^2 + \sigma_\theta^2}{\sigma_{t-1,KF}^2 + \sigma_\theta^2 + \phi^2} \tag{8}$$

$K_t \in [0,1]$ (Kalman gain) identifies the relative contribution of the new evidence $b_t$ in the updated belief.

### 3.4. Survey descriptions

To gain a deeper understanding of physicians' judgment and decision-making regarding HTN and especially to validate modeling constructs regarding the learning and decision-making modules, we designed a seven-question survey. The survey was based on two key assumptions: first, that clinicians measure BP during office visits, and second, that the patients are returning patients. Each question was crafted to understand how physicians incorporate any elements of our model into their judgment and decision-making process (Appendix A). Questions 1 and 2 were designed to validate whether physicians have prior expectations or beliefs about their patient's BP. Question 3 was designed to explore whether physicians consider both past and present patient BP readings. It explores the dynamic aspect of decision-making in HTN management, where both historical and current data play a role. Question 4 was specifically designed to explore and validate the concept of surprise among physicians. In our model, surprise represents a significant deviation in the patient's current BP from the physician's expectation, potentially signaling a change in the patient's BP, which may trigger treatment change. Questions 5–6 were designed to further investigate the concept of $\Delta$ in determining the surprise occurrence. Finally, we asked Question 7 to understand how physicians respond to surprise. This question sheds light on the actions physicians take to determine if these deviations indicate a real change in the patient's BP or are merely random fluctuations. This is a crucial aspect of medical decision-making, as it can influence the timing and nature of interventions and, ultimately, patient outcomes.

## 4. Decision-Making for Medication Prescription

### 4.1. Optimization Framework

In this section, we present the MDP models designed to optimize antihypertensive prescriptions. We have a distinct MDP for each learning strategy, specifically SIL and KF, and another MDP that serves as a benchmark for decision-making under perfect information, denoted as PI. The objective of the MDPs is to optimize prescription decisions to maximize patients' total QALYs, considering the risks of CVD and death, along with medication effectiveness (i.e., reducing BP) and side effects (i.e., reducing quality of life). For model tractability, we assume that the physician will prescribe a standard medication dosage (Law et al. 2009), and the patients fully comply with the prescribed medications. Let $\mathbf{MDP}^{\jmath}$, where $\jmath \in \{PI, KF, SIL\}$ denote the three optimization models mentioned above. To avoid repetition, we present the common components of all three optimization models with a focus on $\jmath = SIL$ and provide the key differences whenever needed.

**Decision epochs**: In HTN management practices, clinical BP measurements and treatment decisions are performed periodically, often quarterly. We denote the decision epochs by $t = 0,1,2,\dots,T < \infty$, representing the number of quarters above the age of 40, the typical starting age for HTN control (Lovibond et al. 2011).

**State space**: We define $x_t = (y_t, \boldsymbol{m}_t)$ as the state variable, where $y_t$ and $\boldsymbol{m}_t$ correspond to the patient's health and medication states. The health state depends on whether the physician has perfect information about the patient's BP, and, if not, which strategy they adopt to learn about the patient's BP. Therefore, we use index $\jmath$ for the state variable, i.e., $x_t^{\jmath} = \left(y_t^{\jmath}, \boldsymbol{m}_t\right)$. We also define three special states $\mathcal{E} = \{E_1, E_2, E_3\}$ as the terminal health states, where $E_1$ and $E_2$correspond to coronary heart disease (CHD) and stroke (ST), representing the most severe types of CVD, and $E_3$ represents the non-CVD mortality event. A patient who faces a terminal state or reaches decision epoch $T$ (whichever earlier) exits the process and receives a lump-sum reward.

Now, we characterize $y_t^{\jmath}$ for the three MDPs. Under perfect information, where $\theta_t$ can be directly observed, we have $y_t^{PI} = h_t^{PI}$: $h_t^{PI} \in \{\mathcal{B}^{PI} \cup \mathcal{E}\}$, where $\mathcal{B}^{PI} = \left\{\underline{B}, \underline{B} + \Delta B^{PI}, \dots, \bar{B}\right\}$ with elements $\theta_t$ representing the discrete BP values between minimum $\underline{B}$ and maximum $\bar{B}$ on the grid size $\Delta B^{PI}$. Under noisy BP observations, if the KF strategy is used, $\mu_t^{KF}$ is sufficient to characterize the patient's health state; hence $y_t^{KF} = h_t^{KF}$: $h_t^{KF} \in \{\mathcal{B}^{KF} \cup \mathcal{E}\}$ , where $\mathcal{B}^{KF} = \left\{\underline{B}, \underline{B} + \Delta B^{KF}, \dots, \bar{B}\right\}$ represents discrete BP values for $\mu_t^{KF}$. Finally, under SIL, in addition to $\mu_t^{SIL}$, we need the number of consecutive non-surprising periods, denoted by $n_t$, to fully characterize the state variable; therefore $y_t^{SIL} = (h_t^{SIL}, n_t) : h_t^{SIL} \in \{\mathcal{B}^{SIL} \cup \mathcal{E}\}$, where $\mathcal{B}^{SIL} = \left\{\underline{B}, \underline{B} + \Delta B^{SIL}, \dots, \bar{B}\right\}$ represents discrete BP values for $\mu_t^{SIL}$ and $0 \le n_t \le N$. Thus, $n_t = 0$ indicates a visit at which a surprise has occurred ($\mathscr{s}_t = 1$), and $n_t \neq 0$ indicates a visit with non-surprising observation ($\mathscr{s}_t = 0$). Thus, $n_t = N$ designates patient's BP stability over $N$ periods (because no surprise has occurred for $N$ consecutive periods). Hence, $n_t$ captures not only the surprise status of the belief regarding the underlying BP regimen $\theta_t$

but also the degree of certainty associated with such belief because of its one-to-one inverse relationship with $\sigma^2_{t,SIL}$. We can also interpret $n_t$ as the *belief strength* (or confidence) under the SIL strategy. Intuitively, $n_t = 0$ indicates minimum confidence because it represents surprise. We take $n_t = N$ as a terminal state to ensure that $n_t$ does not grow indefinitely. Under each optimization model, we denote the non-terminal states by $\hat{x}^{j}_t = (\hat{y}^{j}_t, \boldsymbol{m}_t)$. For example, $\hat{y}^{SIL}_t = (\hat{h}^{SIL}_t, n_t)$; where $\hat{h}^{SIL}_t = \mu^{SIL}_t \in \mathcal{B}^{SIL}$ *and* $n_t \neq N$.

The second component of the state variable, i.e., the patient's medication state, is represented by the vector $\boldsymbol{m}_t = (m_{1t}, m_{2t}, \dots, m_{Mt})$, which includes information regarding the list of antihypertensive medications the patient is taking by $t$. Therefore, $m_{it} = \{0,1\}$ indicates whether the patient is taking medication $i$ in $t$, and $M$ is the total number of medications on the physician's list; hence, $M = 5$. The medications can be ordered in ascending order based on their aggressiveness, meaning that the more effective a medication is, the greater its side effects are likely to be. Let $\{1,2,\dots,M\}$ be the set of medications ordered accordingly.

**Action space**: At each decision epoch and given state $x^{j}_t$, the physician decides if and which medication should be added to the existing list of medications taken by the patient. We impose two restrictions on these decisions following the common clinical practice. First, at each visit, a maximum of one medication can be added to the drugs the patient is taking (NIHCE 2011). Second, when a medication is prescribed, it cannot be removed afterward because of adverse consequences associated with withdrawal, e.g., rebound HTN, increased risk of CVD, and death (Leung et al. 2017, Reidenberg 2011, Williams et al. 2018). In summary, the action space $\mathcal{A}_t(x^{j}_t)$ with elements $a_t(x^{j}_t)$ is defined as follows:

$$\mathcal{A}_t(x^{j}_t) = \{0, \mathcal{I}_t\} \tag{9}$$

where $\mathcal{I}_t = \{i: m_{it} = 0\}$ indicates the set of unprescribed medications. Thus, the feasible action at each decision epoch is either prescribing one of the remaining medications on the list ($a_t(x^{j}_t) = i$) or doing nothing ($a_t(x^{j}_t) = 0$). To simplify the notations, we eliminate the dependency on the state and simply use $a_t$ and $\mathcal{A}_t$.

We operationalize the effects of aging and the medication prescription on BP by shifting its mean. To this end, we define $\eta$ and $\epsilon_i$ as the average increase in the BP mean as a result of a one-year increase in the patient's age (Yano 2017) and the average decrease in the patient's BP as a result of taking medication $i$ (Law et al. 2009). By substituting everything into Eq. (4), the surprise is defined as follows:

$$s_t = \begin{cases} 1, & \text{if } \left|\left(\mu_{t-1} + \eta\Delta t - \sum_i \epsilon_i \mathbb{I}_{(a_{t-1}=i)}\right) - b_t\right| \geq \Delta \\ 0, & \text{otherwise} \end{cases} \tag{10}$$

where $\mathbb{I}_{statement}$ is the indicator function that equals 1 if the statement holds true; otherwise, it equals 0. Note that Eq. (12) aligns with our definition of surprise: any unexpected and significantly large change in the BP (greater than $\Delta$) that cannot be attributed to changes in medication status or the patient's age.

**Transition functions**: Given the current state $x^{j}_t$, the transition function into a new state $x^{j}_{t+1}$ depends on the physician's learning strategy and can be fully characterized using medication decision $a_t$, the conditional probability of observing a new BP measurement $b_{t+1}$, and the probability of entering one of the three terminal states in $\mathcal{E} = \{E_1, E_2, E_3\}$. Appendix C presents the expression of the state transition function.

**Intermediate reward function:** The intermediate reward function $\bar{r}_t(\hat{x}_t^{\jmath}, a_t)$ quantifies one-period QALYs of a patient who is in a non-terminal state $\hat{x}_t^{\jmath}$ and follows decision $a_t$. To incorporate the impact of medication side effects, we assume that medication $i$ reduces QALYs by $\delta_i$, which captures the decrements in quality of life caused by the medication. Assuming that the patient will not encounter any terminal event by $t+1$, the patient's event-free intermediate reward function is:

$$\hat{r}_t(\hat{x}_t^{\jmath}, a_t) = \Delta t\left(1 - \sum_i \delta_i m_{i(t+1)}\right) \tag{11}$$

where $m_{i(t+1)}$ is the $i^{th}$ element of $\boldsymbol{m}_{t+1}$. To consider the possibility of facing terminal events by the next period, we prorate the above QALYs based on half-cycle correction (Sonnenberg and Beck 1993) and adjust the calculation of QALYs, assuming that the terminal events will occur in the middle of the period. Let $w_j$ be the QALYs of a patient who experiences event $E_j$ during period $t$, where $j \in \{1,2,3\}$ (hence, $w_3 = 0$, corresponding to death). Then, the expected QALYs over each period can be expressed as follows:

$$Q_j = \left(\hat{r}_t(\hat{x}_t^{\jmath}, a_t) + w_j\right)/2 \tag{12}$$

We can now present the expected intermediate reward function as follows:

$$\bar{r}_t(\hat{x}_t^{\jmath}, a_t) = \left(1 - \sum_{j=1}^{3} \bar{\alpha}'_{jt}(\hat{x}_t^{\jmath}, a_t)\right)\hat{r}_t(\hat{x}_t^{\jmath}, a_t) + \sum_{j=1}^{3} \bar{\alpha}'_{jt}(\hat{x}_t^{\jmath}, a_t) Q_j \tag{13}$$

where $\bar{\alpha}'_{jt}(\hat{x}_t^{\jmath}, a_t)$ denotes the transition probability from non-terminal state $\hat{x}_t^{\jmath}$ to one of the three terminal states $E_j, j = 1,2,3$. We customize the estimations of $\bar{\alpha}'_{jt}(\hat{x}_t^{\jmath}, a_t)$ to the patient characteristics (Appendix F).

**Lump-sum terminal reward function:** We denote by $R_{jt}^{(1)}$ and $R_t^{(2)}$, respectively, the total expected terminal rewards for a patient in the terminal states $x_t^{\jmath} = E_j$ or $n_t = N$ (when $\jmath = SIL$). More specifically, $R_{jt}^{(1)}$ is the quality-adjusted life expectancy (QALE) of a patient who experiences the first non-fatal event or death in $t$. Note that $R_{jt}^{(1)}$ captures the quality and quantity of the expected life after event $j$, (therefore, $R_{3t}^{(1)} = 0$). Moreover, $R_t^{(2)}$ captures the QALE of a patient who reaches $n_t = N$, indicating the patient's BP stability.

Combining all the components, we can now formulate the value function for **MDP**$^{\jmath}$. Let $v_t(\hat{x}_t^{\jmath})$ be the maximum expected total discounted QALYs earned by a patient given the non-terminal state $\hat{x}_t^{\jmath}$ under learning strategy $\jmath$. The Bellman optimality equations for the non-terminal states $\hat{x}_t^{\jmath}$ can be written as:

$$v_t^{\jmath}(\hat{x}_t^{\jmath}) = \max_{a_t \in \mathcal{A}_t}\left\{\bar{r}_t(\hat{x}_t^{\jmath}, a_t) + \lambda^{\Delta t}\sum_{\hat{x}_{t+1}^{\jmath}} p_t(\hat{x}_{t+1}^{\jmath} | \hat{x}_t^{\jmath}, a_t) v_{t+1}(\hat{x}_{t+1}^{\jmath})\right\} \tag{14}$$

$$\forall t = 0, \dots, T-1;\ \hat{x}_t^{\jmath}: \text{non-terminal}$$

where $\lambda \in [0,1]$ denotes the annual discount factor, and $\Delta t$ is the time difference between two consecutive decision epochs. For terminal states, we define the value functions as follows:

$$v_t^{\jmath}(E_j, \boldsymbol{m}_t) = R_{jt}^{(1)};\ v_t^{SIL}(\hat{x}_t^{SIL}, n_t = N, \boldsymbol{m}_t) = R_t^{(2)};\ v_t^{\jmath}(x_T^{\jmath}) = R_T^{(3)} \\ \forall x_T^{\jmath};\ \forall t = 0, \dots, T-1 \tag{15}$$

where $R_T^{(3)}$ is the remaining QALE of a patient at the end of the decision horizon $T$.

Recall that the SIL mechanism captures how clinicians reconcile new information with their prior expectations, manifesting the complex dynamics of clinical inertia based on surprise. Therefore, $\mathbf{MDP}^{SIL}$ formulates the trade-off between exploration (delaying treatment to gather more information) and exploitation (acting on available information). This trade-off is represented by the state variable $n$, which quantifies the number of visits without surprise, which also captures the physician's confidence in their beliefs about the patient's BP. A surprise ($n$=0) represents maximum uncertainty, while increasing $n$ reflects growing confidence in the beliefs. This dynamic is integral to understanding decision-making under clinical inertia, where physicians delay action to seek greater certainty, risking delays in necessary interventions. Our framework provides a novel analytical approach to addressing this pervasive issue, directly tackling what Ogedegbe (2008) describes as the "*wait until next visit*" approach, a key physician-related barrier to optimal HTN control. Importantly, our study aligns with the MAP (*Measure Accurately, Act Rapidly,* and Partner With Patients) protocol (Egan et al. 2018), which emphasizes the importance of timely actions to counter clinical inertia. By explicitly incorporating the dynamics of BP uncertainty and physician's judgment bias into the decision-making process, our framework not only identifies when delays are necessary for gathering evidence but also determines when they become suboptimal, thereby mitigating the risks associated with prolonged untreated HTN.

### 4.2. Structural Properties of the Optimal Policy Under Physician Judgment

We present the structural properties of the optimal policies and delegate the technical details to Appendix D to provide conditions for the optimality of the threshold policy and its implications in practice. The theoretical results will be illustrated for the models calibrated based on real-life data (§5.1) and the medical literature. The details of parameter estimations for model calibration are in Appendix F. Although threshold policy is optimal for all above MDPs, we will focus on $\mathbf{MDP}^{SIL}$, because its state space is more complex.

We establish that there exist a minimum BP level and a minimum belief strength level for prescribing medications. The common approach to characterize optimality of the threshold policy includes proof that the value function is monotone in the states and the action-value function has monotone differences in state-action pairs, a proof that these properties are preserved under the MDP operator, and an induction-based argument to show that the desired structural properties hold for all decision epochs (Puterman 2005). We adapt the approach for $\mathbf{MDP}^{SIL}$ to account for its additional state $n_t$ compared to the unidimensional state space for $\mathbf{MDP}^{KF}$ and $\mathbf{MDP}^{PI}$. The structural results hold only for SIL under the optimal $\Delta$ (explained in §5.3).

**Theorem 1.** *Under Assumptions 1–4 in Appendix D, the following hold true:*

1. *In each period $t$, medication state $\mathbf{m}_t$, and belief strength $n_t$, there exists a BP threshold $\mu_t^*$ such that:*

$$a_t^*(\hat{x}_t^{SIL}) = \begin{cases} 0, & \mu_t < \mu_t^*(n_t) \\ i, & \mu_t \geq \mu_t^*(n_t) \end{cases} \tag{16}$$

2. *In each period $t$, medication state $\mathbf{m}_t$, and BP state $\mu_t$, there exists a belief strength threshold $n_t^*$ such that:*

$$a_t^*(\hat{x}_t^{SIL}) = \begin{cases} 0, & n_t < n_t^*(\mu_t) \\ i, & n_t \geq n_t^*(\mu_t) \end{cases} \tag{17}$$

**Theorem 2.** *Under Assumptions 1-4 in Appendix D, $\mu_t^*(n_t)$ is (1) nonincreasing with $n_t$ and (2) nonincreasing with $t$.*

Theorem 1 indicates that there is a minimum threshold for the estimated BP level and its corresponding belief strength in making decisions about prescriptions. According to these Theorems (not shown for $a_t^*$ with $t$), the thresholds exhibit pairwise inverse relationships between $\mu_t$, $n_t$, and $t$, which means that as one state increases, the threshold for another state decreases, suggesting important clinical implications. First, the threshold policies concerning $\mu_t$ and $n_t$ suggest that the lower the belief strength, the higher the minimum level of BP state to prescribe medication and vice-versa. From the perspective of clinical inertia, these structural properties provide actionable insights into the impact of BP uncertainty on the optimal prescription thresholds. This is because clinical inertia, characterized by delayed treatment initiation or adjustment, often stems from uncertainty and the resulting low confidence in BP status. The thresholds reveal that when confidence is low (low n), higher BP levels (higher $\mu$) are required to prescribe medications. This delay avoids unnecessary interventions unless the BP is excessively high. Conversely, increased confidence (high *n*) enables physicians to act decisively at lower BP levels, mitigating inertia and promoting timely interventions. This mitigates the risk of delayed treatment by facilitating timely interventions in line with evidence. While inertia can lead to suboptimal outcomes due to delayed action, it also serves a protective role by preventing premature decisions based on noisy or unreliable measurements. These findings underscore the importance of enhancing physician confidence through accurate measurements and robust learning mechanisms. While delay (i.e., inertia) is often viewed negatively, our analysis demonstrates that not all delays are suboptimal; however, the key question is when a delay represents the optimal decision and when it does not. Our decision analytics framework makes such distinctions, providing clinically meaningful tools and insights. The derived thresholds indicate that delay can be an optimal strategy when additional information is required to mitigate uncertainty, thereby enabling evidence-based interventions. Conversely, delays become suboptimal when they prolong high BP conditions unnecessarily, compromising patient outcomes. These findings advance our understanding of clinical inertia, offering actionable strategies for balancing caution with timely action.

The 2018 ESC/ESH HTN guideline by the European Heart Association has highlighted the inverse relationship between BP severity (equivalent to our $\mu_t^*$) and the number of office visits (equivalent to our $n_t^*$) (Williams et al. 2018). Similarly, the higher the BP level, the lower the minimum belief strength needed for prescribing a medication, and vice versa. Second, the BP threshold for medication prescription does not increase with age. Although controversial in the medical community (Myers et al. 2015), this observation is more consistent with important studies, such as the well-known SPRINT study (SPRINT Research Group 2015) and an empirical study based on the observations of 8.8 million patients in the US (Fletcher et al. 2017). Both studies indicate that older adults should maintain a lower BP than younger adults. Moreover, in a recent 5-year longitudinal study of 3,627 community-dwelling individuals aged 65 years or older, the presence of an elevated BP was associated with a significant increase in the CVD risks (Myers et al. 2015). Jaeger et al. (2022) argue that the benefit from the HTN treatment increases with age as aging increases CVD risk.

## 5. Empirical Findings and Numerical Insights

First, we describe a carefully designed clinical protocol we used to measure BP accurately and collect our data for model calibration. After briefly presenting the findings of our survey, we illustrate optimal policies under $\mathbf{MDP}^{SIL}$. Next, we present a series of extensive numerical analyses to examine DQD comprehensively.

### 5.1. Clinical Setting and Data Description

We calibrated our models using primary data collected in a unique clinical setting, where the same cohort of patients underwent both conventional and new BP measurement technologies, with measurements were obtained both in short and long intervals. Our primary data source for the parameter estimation and model calibration includes two real-life datasets collected from 376 patients over six years at the Vascular Health Unit in Montreal General Hospital in Canada. For the parameters that could not be estimated from our data, we consulted medical literature to develop a complete set of realistic parameters for our numerical analyses.

Our two real-life datasets were collected through carefully designed clinical BP measurement settings. In the first clinical setting, at each visit, a rigorous protocol was followed to measure BP using the AOBP technology (Daskalopoulou et al. 2015, Leung et al. 2017). We use this measurement as a reasonable proxy for the underlying BP (i.e., ground truth) and its long-term variability. This is a standard clinical setting for measuring LBPV in clinical studies (Mehlum et al. 2018). Following the protocol, after an appropriate resting period (3-5 minutes), six consecutive measurements of the patient's BP are taken within 1-3 minutes intervals in a private examining room. In the second setting, the same patients were asked to wear a device to measure their BP every 20–30 minutes for 24–48 hours under their living conditions. This ambulatory BP monitoring (ABPM) provides us with SBPV as well as the noise around the true underlying BP captured in the first setting. This is a standard setting for measuring SBPV in clinical studies (Mehlum et al. 2018). Because the measurements in the first dataset were made during the day when the two datasets had to be linked, we filtered the second dataset to the daytime records of the same patients obtained within one week of their AOBP assessments. The details of the data sources and the parameter estimation procedure are in Appendix F.

### 5.2. Survey Findings

In our survey, we received responses from 25 physicians with various specialties. 68% of the respondents identified as male, over 50% were aged 40 or older, and 72% had more than 10 years of experience in their respective fields. The participants represented a diverse range of medical specialties, with more than 14 specialties reported. This diversity in the participant pool enhances the generalizability of our survey results, as it captures a wide range of perspectives and experiences in medicine.

One of the key findings from our survey was the significant role of historical data in decision-making. When asked '*After measuring the patient's BP, do your decisions regarding BP management (e.g., changing their antihypertensive medications or requesting more BP information) depend on past BP measures, current visit measures, or both?*', a substantial majority of physicians (88%) reported that they consider both past and current visit BP measures when making decisions for HTN management. This finding underscores the importance of incorporating both historical and current information, reflecting the real-world decision-making processes. Most respondents (73%) reported being sensitive to unexpected BP measurements in their decision-making, validating the inclusion of surprise in our model. When asked '*If my patient's BP measurement in the current visit is significantly different from my expectation regarding their BP status, what is your reaction?*', 73% of physicians responded that such a discrepancy would alert them about an unexpected observation about the patient's BP, leading to a change in their decisions. It is notable that within that 73%, a substantial majority (63.6%) had over 10 years of experience in their field. In response to the question regarding the significance of a difference between a patient's current BP and the physician's expectations, 71.4% of physicians (61.9% with more than 10 years of experience) indicated that they consider such a difference significant if it exceeds a certain threshold. This finding supports the inclusion of $\Delta$ in our model, representing the threshold for determining a surprise. Other details of the survey are available in Appendix A.

### 5.3. Illustration of Optimal Medication Prescription Policy

After model calibration (Appendix F), we solved $\mathbf{MDP}^{SIL}$ using monotone backward induction (Puterman 2005). We present the numerical results for a risk-free female (non-smoker and non-diabetic with normal blood cholesterol) under moderate levels of LBPV, SBPV, and device noise. However, for the empirical analysis in §5.4.1, we included data from males to examine the impact of sex and increase the sample size.

Fig 3-1 depicts a 3D plot illustrating optimal thresholds associated with the least aggressive medication, i.e., ACEI, as a first-line treatment. To demonstrate how the optimal policy changes in relation to belief strength $n_t$, which also signifies the number of consecutive non-surprising visits, and age $t$, we have included cross-sections of the thresholds for ACEI in Figs 3-2 to 3-4. As seen in Fig 3-1, the threshold for ACEI decreases in $t$ and $n_t$. This suggests that it is optimal to start with a high BP threshold to prescribe ACEI only if the physician's best estimate (belief) about the underlying BP exceeds the threshold and gradually lower the threshold as the patient gets older and/or the physician gains more confidence in their beliefs about the patient's underlying BP (Hazra et al. 2019). Figs 3-2 and 3-3 show that the ACEI threshold initially decreases sharply with $n_t$ and stabilizes around $n_t = 15$, whereas it decreases almost linearly with age. Concerning aging, it is optimal to adjust the threshold in a linear fashion in order to account for the effect of aging on BP at a constantly deteriorating rate. However, as $n_t$ increases, the impact of new BP observations diminishes, resulting in stabilized beliefs about true BP, hence a steady threshold for medication prescription. The concave shape and downward shifting contours of the ACEI thresholds in $t$ and $n_t$, as shown in Fig 3-2, further support these observations. Fig 3-3 shows that the lowest threshold for prescribing ACEI as the first-line treatment (~140 mmHg) is advised when both belief strength and age are highest. Moreover, the widest range for prescribing ACEI is between 140 and 155 (Fig A-4). Figs 4-1 and 4-2 illustrate optimal thresholds associated with all five medication classes for the same patient.

The illustrated results reveal that when confidence is low (low n), higher BP thresholds (u), hence higher physicians' hesitancy to act are recommended, delaying treatment initiation or adjustment to gather more evidence and avoid unnecessary interventions. However, when BP levels are excessively high, immediate intervention (i.e., avoiding hesitancy) is warranted despite low confidence, prioritizing patient safety. On the other hand, as confidence increases (high n), lower BP thresholds for prescribing medication are sufficient, allowing for timely decisions and reducing the risks associated with sustained high BP.

When comparing the optimal thresholds for different medications, we notice that they are ordered from least (bottom) to most aggressive. This is consistent with the medical guidelines, which recommend that the medication aggressiveness must be increased gradually (Boffa et al. 2019, Leung et al. 2017, McCarthy et al. 2024, NICE 2015). This does not mean physicians must always start with the least aggressive medication. In fact, our results indicate that, depending on the BP level, any of the five medications could be a candidate for the optimal first-line treatment. For example, if the best estimate of the underlying BP is 182 mmHg based on $n_t = 10$ consecutive non-surprising observations, the first-line treatment must be the most aggressive medication (i.e., $i = 5$). This is a departure from the existing guidelines that recommend the same first-line treatment (here, the least aggressive medication $i = 1$) regardless of BP level (James et al. 2014).

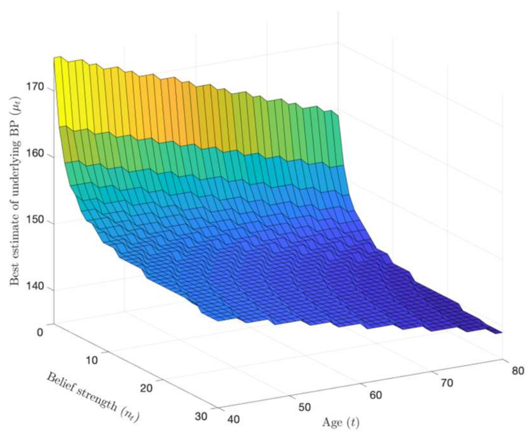


**Fig 3-1** Threshold for ACEI with respect to $\mu_t, n_t$ and $t$.

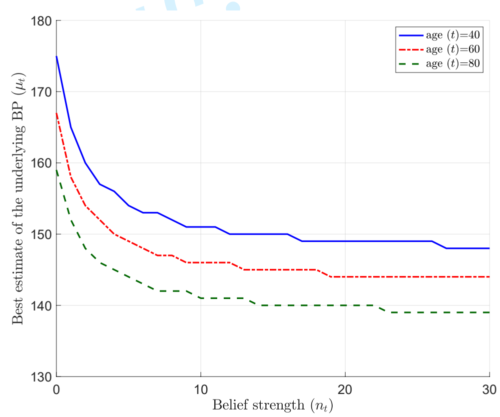


**Fig 3-2** Threshold for ACEI with respect to $\mu_t$ and $n_t$ for $t$= 40, 60, and 80

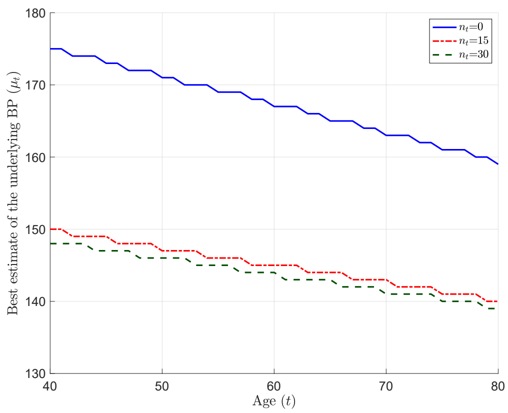


**Fig 3-3** Threshold for ACEI with respect to $\mu_t$ and $t$ for $n_t$=0,15, and 30

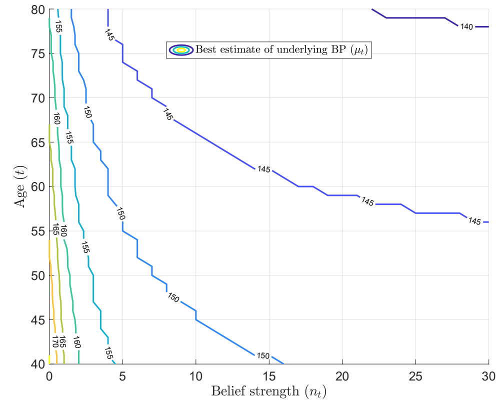


**Fig 3-4** Contour plots of the threshold for ACEI

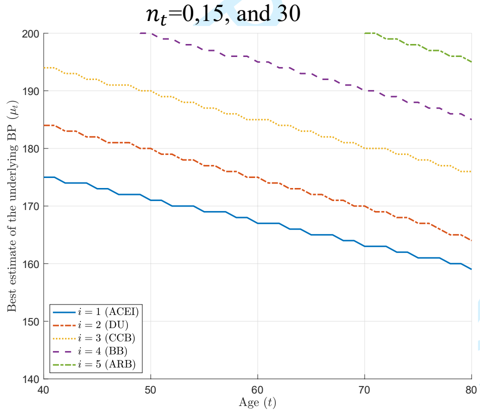


**Fig 4-1** Thresholds for all medications with respect to $t$ for $n_t = 0$

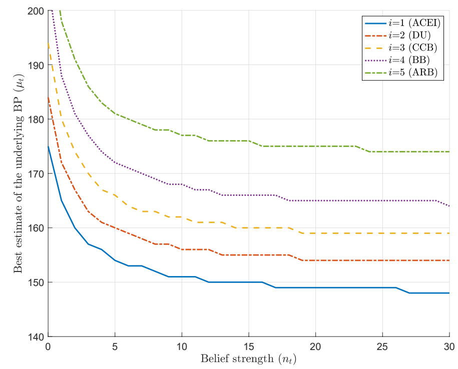


**Fig 4-2** Thresholds for all medications with respect to $n_t$ for $t = 40$

### 5.4. Decision Quality Decrement

In this section, we analyze DQD, as a key performance indicator (KPI), to understand the extent to which decision quality, in terms of the patient's QALY, degrades due to HF (physician judgment) and TF (BP measurement noise). To this end, we define the DQD under learning strategy $\jmath$ as $DQD^{\jmath} = v^{PI} - v^{\jmath}$, where $v^{PI}$ represents the maximum expected payoff under accurate measurement protocol with new technology, which provides almost perfect information regarding underlying BP, and $v^{\jmath}$ represents the maximum expected payoff under the $\jmath$-type learning strategy. The maximum expected payoff of each scenario is the quantification of the value functions obtained from the corresponding MDPs. Furthermore, we can decompose DQD into the two following components (see Fig 1):

$$DQD^{SIL(\Delta)} = \underbrace{v^{PI} - v^{KF}}_{DQD_{TF}} + \underbrace{v^{KF} - v^{SIL(\Delta)}}_{DQD_{HF}} \tag{18}$$

The first term on the right-hand side represents the loss in QALYs due to TF, noting that KF is free from human bias. The second term captures the loss in QALYs attributable to the HF, reflecting the physician's judgment bias, such as over-reaction or -reaction to the patient's BP. This decomposition allows us to separately evaluate the impacts of each factor on decision quality. The degree of bias under SIL depends on the value of $\Delta$; therefore, we analyze DQD under three different $\Delta$ levels, i.e., $\Delta^L = 0$, $\Delta^*$, and $\Delta^H = \infty$. The first and third levels correspond to extreme cases. Under $\Delta^L$, the physician relies solely on the most recent observation, disregarding prior history. Conversely, under $\Delta^{H}$, which is the same as the standard Bayesian updating mechanism, the physician gradually converges to a constant belief by undervaluing new BP readings due to overcommitment to beliefs. The case of $\Delta^{*}$ represents the optimal commitment to belief, hence a minimum judgment bias. It is defined to minimize the mean of squared errors in estimating the underlying BP regimen (please refer to Appendix F for the definition and calculations of $\Delta^{*}$). We normalize DQD by calculating the following ratios (Ketzenberg et al. 2015):

$$NDQD^{\hat{\jmath}} = (v^{PI} - v^{\hat{\jmath}})/v^{\hat{\jmath}} \tag{19}$$

$NDQD^{\hat{\jmath}}$ indicates relative decrease in patient's expected QALYs resulting from imperfect information under learning strategy $\hat{\jmath}$. As a KPI, this measure enables us to quantify the relative effects of HF and TF on decision quality, serving as a direct indicator of the efficiency loss in patient care. In our study, DQD is influenced not only by the physician's learning strategy and their degree of bias but also by patient characteristics, sources of BP uncertainty (i.e., $\sigma_{\theta}$, $\sigma_{b}$, and $\tau$), and decision flexibility, which will be analyzed next.

### 5.4.1. Main and Interactive Determinants of QDQ

To explore the impact of various factors on DQD, we conducted a pairwise full factorial analysis of their main and interactive effects on DQD. These factors were:

- **F1:** *HF (physician judgment bias):* assessed through $\Delta$ at three levels: low ($L$), moderate ($M$), and high ($H$).
- **F2:** *TF (measurement noise):* assessed through $\tau$ at three levels: $L$, $M$, and $H$.
- **F3:** *Patient's LBPV:* assessed through $\sigma_{\theta}$ at three levels: $L$, $M$, and $H$.
- **F4:** *Patient's SBPV:* assessed through $\sigma_{b}$ at three levels: $L$, $M$, and $H$.
- **F5:** *Patient's sex:* Male and Female.

To this end, we derived 162 optimization scenarios (encompassing all combinations of the factor levels) from the MDP value function quantifications to prepare the metadata for analysis. We employed Analysis of Variance (ANOVA) to examine the interactions between the factors in two ways, ensuring the robustness of the analyses:

- **Original Data Analysis**: We performed ANOVA on the original metadata consisting of 162 records. We calculated the p-value of the statistical significance of the main (i.e., individual) effect of each factor and its pairwise interaction with the other four factors.
- **Bootstrapping**: To ensure the robustness of the ANOVA results, we performed bootstrapping with 3,000 iterations by repeatedly sampling from the original metadata. We calculated the p-value of the statistical significance of the effects (main and interactive) and the percentage of times (out of 3,000 iterations) the p-value was less than 0.05.

Our empirical investigation of the interactive effects of various factors on DQD reveals several significant interactions that affect patient outcomes in terms of QALY (Table 1). These findings indicate the complex dynamics between HF, TF, and even patient characteristics, offering critical insights into how these interactions can guide more informed clinical and policy decisions.

**Table 1** Main and interactive effects of various factors on DQD

| F1 (HF) | | F2 (TF) | | F3 (LBPV) | | F4 (SBPV) | | F5 (Sex) | |
|---|---|---|---|---|---|---|---|---|---|
| Original | Boot | Original | Boot | Original | Boot | Original | Boot | Original | Boot |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Interaction Effects*** | | | | | | | | | | |
| **F1** | | | **0.0113** | **0.0195 (89.1%)** | **0.0000** | **0.0005 (99.8%)** | 0.8459 | 0.3810 (12.9%) | **0.0092** | **0.0367 (82%)** |
| **F2** | | | | | 1.0000 | 0.5084 (6%) | 0.9078 | 0.4165 (11.2%) | **0.0371** | **0.0535 (77%)** |
| **F3** | | | | | | | 1.0000 | 0.5164 (5.4%) | 1.0000 | 0.5113 (5.4%) |
| **F4** | | | | | | | | | 1.0000 | 0.5018 (5.4%) |
| **Main Effects**** | | | | | | | | | | |
| | 4.6e-22 | 3.5e-15 | 3.5e-13 | 4.6e-09 | 6.9e-07 | 5.5e-04 | 1.6e-04 | 4.6e-03 | 2.1e-05 | 4.5e-03 |

**Notes:** ***Boot:*** bootstrapping;
**Under "**Original**," the figures indicate the p-value of the statistical significance of the interaction with the original data (i.e., without bootstrapping). Under "**Boot**," the figures without brackets indicate the p-value of the statistical significance of the interaction with bootstrapping, and the figures in the brackets indicate the percentage of times (among 3,000 iterations) where the p-value of the statistical significance of the interaction was lower than 5%. Statistically significant values are **boldfaced**.*
***The figures indicate the p-value of the statistical significance*

Our results indicated that the interaction between physician judgment and technology was significant in both the original metadata and the bootstrapping analyses (p-values: 0.0113 and 0.0195, respectively), suggesting that the impact of physician judgment on decision quality relies on technology and vice versa. The robustness of this interaction, indicated by its consistent significance across 89.1% of bootstrap interactions, emphasizes the intertwined nature of these factors. The effect size of this interaction, as measured by partial eta squared, indicated a moderate effect of 0.11. This measure reflects the proportion of the total variance in the dependent variable that can be attributed to the interaction term after accounting for the variance explained by other terms in the model. In the context of ANOVA, effect sizes in this range are considered meaningful and indicate that the interaction between HF and TF has a substantial impact on DQD. To further ensure the robustness of our analyses, we also performed the Kruskal-Wallis test, followed by a post-hoc exploration of the interaction significance. This non-parametric alternative to the one-way ANOVA is insensitive to the ANOVA assumptions. Since it does not directly test for interactions between factors, we first conducted the test for each factor independently, followed by post-hoc pairwise comparisons. This involved creating combined groups to represent each interaction level and then performing the Kruskal-Wallis test on the combined groups. The results indicate significant main effects for both factors (p-values: 5.59e-16 for HF and 2.25e-08 for TF) and their interaction (p-value:1.27e-09), supporting the significance of this interaction in explaining the variability in DQD. This finding supports our earlier assertion about the dual impact of device measurement error: its direct effect through inaccurate readings and its indirect effects on physician judgment.

Our analysis also revealed a strong interaction between HF and LBPV, with nearly universal significance in bootstrap scenarios (99.8%). This underscores how long-term BP trends significantly affect physician judgments, which may lead to diminished decision quality if these trends are misinterpreted or underestimated. On the other hand, the interaction between HF and SBPV was insignificant, indicating that short-term BP fluctuations have a negligible influence on the physician's decision quality in the presence of their judgment biases. This indicates that immediate BP changes are less likely to impact clinical decisions, which may prioritize more stable, long-term BP information. These results are consistent with existing medical literature highlighting the greater importance of long-term BP variability over short-term fluctuations in HTN management (Gosmanova et al. 2016). It also emphasizes the importance of considering BP variability alongside mean BP, which is often the sole BP measurement used in clinical practice.

The interaction between HF and patient sex was significant, demonstrating that decision quality decrement varies between male and female patients, likely due to physiological and perhaps psychosocial differences that affect treatment outcomes. This interaction was significant in 82% of bootstrap cases, emphasizing the need for gender-sensitive approaches in HTN management. This interaction may reflect either a bias towards sex or the recognition of sex among physicians as a significant risk factor for CVD. Technology's interaction with sex also showed nearly significant results, with a p-value near the threshold of 5% significance in bootstrap analyses (0.0535). This suggests that the impact of technology on decision quality may differ by sex, potentially due to differences in BP patterns or responsiveness to treatment. This interaction might also be influenced by higher-order interactions between human and technology factors, where technology impacts how the physician's judgment affects decision quality. In that case, technological improvements in BP measurements may reduce such biases. Therefore, sex-specific interactions highlight the necessity for personalized and gender-sensitive HTN management strategies from both clinical and equity perspectives. Awareness campaigns emphasizing the significance of consistent BP monitoring through advanced technologies, which not only reduce measurement noise but also lessen judgment biases, could improve patient outcomes and reduce disparities in HTN control. Interestingly, no significant interactions were found between TF and both types of BP variability (long-term and short-term), nor between the BP variability factors themselves. This lack of interaction suggests that BP measurement's impact is generally robust across different patterns of BP variability, whether long-term or short-term.

These findings have profound implications for clinical practice. The significant interactions between HF and TF highlight that reducing judgment biases alone might be insufficient without addressing device errors. While addressing human factors can reduce the negative effects of judgment bias, it does not fully address the technological deficiencies. Conversely, improving technological accuracy through advanced BP measurement technologies can simultaneously enhance measurement precision and reduce judgment bias. This is because one of the primary causes of judgment bias or subjectivity often arises from a lack of trust in the reliability of measurements (E. O'Sullivan and Schofield 2018). Thus, modernizing BP measurement technologies has the dual potential to enhance measurement accuracy and strengthen clinician confidence, resulting in reduced biases that ultimately improve overall decision quality. Moreover, training programs in medical schools that focus on estimating patients' true disease status with uncertain measurements (such as BP and cancer), considering both long-term history and immediate measurements, have great potential to improve patient outcomes. Finally, the significant interactions with sex underline the importance of personalized approaches that account for physiological and potentially psychosocial differences. Tailoring treatment protocols and training programs to these differences could mitigate DQD and improve overall patient outcomes.

#### 5.4.2. Decision Quality Decrement with Respect to the Patient's Baseline Risk

We also analyzed how decision quality decrement changes with respect to the patient's baseline risk levels. In this analysis, we kept the three sources of BP uncertainty (i.e., by $\sigma_\theta$, $\sigma_b$, and $\tau$) at their moderate level. Fig 5 shows the results, where the baseline risk level increases along the horizontal axis. First, decision quality decrement increases in the baseline risk. Note that being male is a risk factor for CVD (D'Agostino et al. 2008). Interestingly, the decrement is higher for diabetic women than for a woman who smokes, whereas the opposite holds for male patients. This observation also agrees with the results from the Framingham risk study (D'Agostino et al. 2008). Second, while the decrement is almost equal under $KF$ and $SIL(\Delta^*)$, it is highest under $SIL(\Delta_H)$. Notably, $SIL(\Delta^*)$ exhibits excellent learning performance, and the new technology would be most beneficial when the physician is overcommitted to their beliefs. Third, while decision quality decrement increases with the patient's risk level, the differences between $SIL(\Delta^*)$ and the two biased learning strategies (i.e., $SIL(\Delta_L)$ and $SIL(\Delta_H)$) remain almost constant, emphasizing that HF is influenced by the degree of BP uncertainty caused by $\sigma_\theta$, $\sigma_b$, and $\tau$.

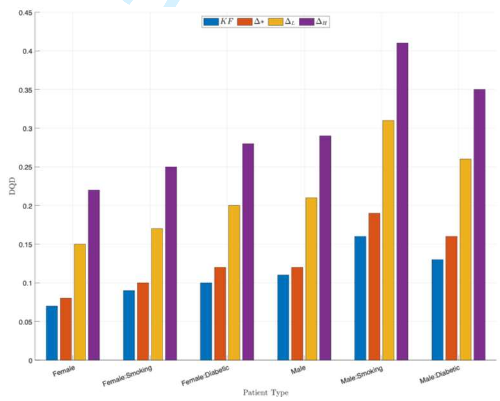


**Fig 5** Changes in the decision quality decrement with respect to patients' characteristics

Our analysis (in Appendix G1) also reveals that DQD increases with greater decision flexibility, defined as the number of medications available for prescription, especially under underestimation bias. This highlights that new technology is most valuable when the full list of medications is available to a physician who is overcommitted to their beliefs. Finally, to further compare the role of the technology versus human factor, we analyzed their shares (i.e., %contributions) in the total DQD (Appendix G2). The results indicate that under $SIL(\Delta^*)$, on average, the human factor has a significantly lower contribution than the technology factor, whereas under $SIL(\Delta_L)$, the two factors are equally important, and under $SIL(\Delta_H)$, the human factor is around 73% more important than the technology factor.

## 6. Conclusions and Implications for Management and Policy

This study provides a comprehensive analysis of HTN management under uncertainty caused by HF (judgment bias) and TF (measurement devise noise) and their impacts on decision quality in terms of patient's quality of life. Judgment bias pertains to the *human elements of error*, an active area of research in healthcare operations management (KC et al. 2020) for its role in the ubiquitous use of heuristics in clinical decision-making. We introduce the notion of surprise to formulate biased learning. The bias in our study is defined as the under-reaction or over-reaction to the observed BP caused by the physicians' over-confidence and under-confidence in their beliefs, respectively. The former is especially important for two reasons: first, the under-reaction bias proxies the physician inertia, which has been recognized in clinical studies, including the European as well as JNC HTN guidelines, as an important barrier to effective HTN management (Gil-Guillen et al. 2010, Williams et al. 2018); second, considering cognitive learning, under-reaction is more common in humans than the overestimation bias (Benjamin 2019), particularly under noisy environments (Mukherjee and Sinha 2018). Our study provides a detailed understanding of the interactive effects of various factors on decision quality in HTN treatment and, hence, patient outcomes. By carefully considering these interactions, particularly those involving human judgment and technology, healthcare providers can enhance the precision and effectiveness of HTN management, ultimately improving patient quality of life. The human factor interacts with all other factors except for SBPV, indicating the critical role of physician judgment in HTN management. Moreover, the impact of human factors on decision quality and patient outcomes is heavily influenced by the technology factor, emphasizing the crucial interplay between these two elements.

**Managerial and policy implications.** Our study offers several key insights with significant implications for healthcare providers, policymakers, and technology developers in the primary care setting. Our research provides a framework for understanding how physicians form and update their beliefs and make decisions under uncertainty caused by their own biases and technology errors. The significant interaction between HF and TF indicates that merely reducing physician judgment biases is insufficient to enhance decision quality without concurrently addressing technological deficiencies. This interaction highlights the need for a dual approach in clinical practice, namely, improving measurement technologies while also mitigating human cognitive biases. Our findings suggest that enhancing the accuracy of these devices not only directly improves measurement reliability but also indirectly reduces the impact of physician biases. This dual benefit arises from the decreased necessity for physicians to overcompensate for perceived measurement errors, thus promoting more objective clinical judgments. Consequently, healthcare policies should prioritize the adoption and integration of state-of-the-art BP measurement technologies, particularly in settings where decision quality is most critical, such as in the management of high-risk patients.

Moreover, our study emphasizes the necessity of incorporating cognitive bias awareness into clinician training programs. By educating healthcare providers on the potential pitfalls of their judgment and how it interacts with technological factors, we can equip them with the tools necessary to make more informed, evidence-based decisions. This training should be complemented by decision-support systems that offer real-time feedback on the quality of decisions and patient outcomes, further mitigating the impact of human errors. The interaction between HF and BP variability reinforces the importance of considering both stable and fluctuating BP trends in clinical practice. Since LBPV significantly impacts decision quality, clinical guidelines should promote more fulsome approaches to BP management that consider more than just mean BP levels. This could involve the development of protocols that integrate LBPV into treatment decisions, ensuring that physicians account for both immediate measurements and longer-term trends.

Finally, the differential impact of biases on patient outcomes based on individual risk profiles, such as cardiovascular risk and BP variability, highlights the importance of personalized medicine. For instance, the interaction between HF and sex reveals that gender-sensitive treatment protocols that consider physiological differences between male and female patients can improve overall patient outcomes. Additionally, technological improvements in BP measurements may reduce gender-related biases by providing more accurate data, thereby supporting more equitable care.

**Limitations and future work**. First, relaxing the standard dose assumption to include half- and twice-standard doses, commonly recommended to physicians, further enhances the applicability of our results. Second, we assume full patient adherence to treatment plans; future work could account for imperfect adherence. Lastly, exploring various cognitive biases in medical decision-making and evaluating interventions to mitigate them could provide actionable insights for healthcare managers and policymakers.

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# ONLINE APPENDIX

## On The Interplay Between Clinical Inertia and Blood Pressure Measurement Technology in Hypertension Management

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## A. Survey Design Complementary

### A.1. Survey Instructions

To gain a deeper understanding of physicians' judgment and decision-making processes regarding BP management and especially to validate modeling constructs regarding the learning and optimization modules in our study, we designed a seven-question survey. In Section A-2, we present the survey questions, which were crafted to understand whether and how physicians incorporate any elements of our model into their judgment and decision-making process in real-world clinical settings. Finally, we present our findings in Section A-3.

### A.2. Survey Questionnaire

**Background**: Thank you very much for responding to our questionnaire. The purpose of this short survey is to gain an understanding about physicians' judgment and decision-making regarding blood pressure (BP) management. This survey is a fundamental part of a more extensive study currently undertaken by researchers from XX Universities[1].
**Note 1:** In this questionnaire, the assumption is that the clinician measures BP *during office visits*. Moreover, the patient is assumed to be a *returning* patient.

**Note 2:** The identity of the surveyed clinician will remain confidential to the study's investigators.

Table A-1: End of Survey Questionnaire

| Question | Response Options |
|---|---|
| **Q1*: Before* measuring my patient's BP:** | A. I have an expectation regarding their current BP, which is based on the patient's past BP records (if available), current medications, and/or all other relevant information (e.g., age, sex, diabetes status, smoking status, etc.).<br>B. I do not have any expectations regarding my patient's BP before measuring their BP during the current visit. |
| **Q2*: Before* measuring my patient's BP:** | A. I have an expected BP target in mind, which is based on the patient's past BP records (if available), current medications, and/or all other |

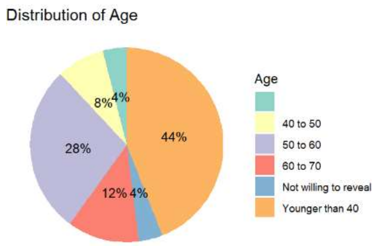
[1] *The names of the institutions have been replaced with XX to maintain anonymity of the co-authors and comply with the double-blind review process.*

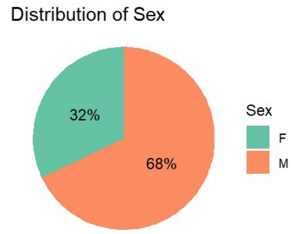
| | |
|---|---|
| | relevant information (e.g., age, sex, diabetes status, smoking status, etc.).<br>B. I do not have a BP target in mind for the current patient before measuring their BP in the current visit. |
| **Q3*: After* measuring the patient's BP, my decisions regarding BP management (e.g., changing their antihypertensive medications or requesting more BP information) depend on:** | A. Both the past BP records (if available) and the current visit's measurement.<br>B. Only the past BP records (if available).<br>C. Only the current visit's BP measurement. |
| **Q4: If my patient's BP measurement in the current visit is significantly different from my *expectation regarding their BP status*:** | A. This alerts me about a rather unexpected observation regarding their BP, leading to a change in their BP management.<br>B. This does not provide me with any actionable information.<br>If the response is B, please skip to Question 7. |
| **Q5: If my patient's BP measurement in the current visit is significantly different from my *expected target*:** | A. This alerts me about a rather unexpected observation regarding their BP, leading to a change in their BP management.<br>B. This does not provide me with any actionable information.<br>If the response is B, please skip to Question 7. |
| **Q6: When there is a difference between the patient's current BP and my expectations:** | A. I will consider this difference *significant* if it is larger than a *threshold level* that depends on my patient's history and demographics. For example, a 10mmHg increase might alert me about an unexpected BP increase for patient X, whereas I could consider the same value an insignificant increase for patient Y.<br>B. I will consider this difference *significant* if it is larger than a *threshold level* that is the same for all my patients.<br>C. I do not pay attention to the extent of this difference. |
| **Q7: If I were to conclude that there is a change in my patient's BP status:** | A. Without needing additional BP measurements, I will immediately make changes to the patient's BP treatment plan, including re-evaluating the antihypertensive drugs.<br>B. To ensure that my patient's unexpected change is not temporary, I will seek more BP measurements (e.g., through home BP monitoring, 24-hr ambulatory BP monitoring, or the measurement at the next visit). Only then will I make changes to the patient's BP management plan, including reassessing their antihypertensive medications.<br>C. I do not do anything. |
| **Sex Identity:** | A. Female; B. Male; C. LGBTQ+; Not willing to reveal |
| **Age** | A. Younger than 40; B. 40–50; C. 50–60; D. 60–70; E. Older than 70; F. Not willing to reveal |
| **Numbers of years of practice** | A. 0–5; B. 5–10; C. More than 10 years; Not willing to reveal |
| **Specialty** | A. Primary care provider/family physician; B. Hypertension Specialist; C. Internist; Endocrinologist; D. Nephrologist; E. Cardiologist; F. Other; G. Not willing to reveal |

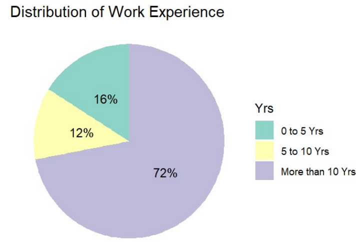
### A.3. Survey Findings

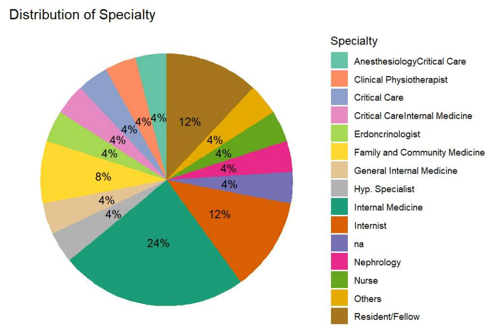
Our survey involved 25 physicians. In what follows, we provide the descriptive statistics for the participants.

**Demographic Information**



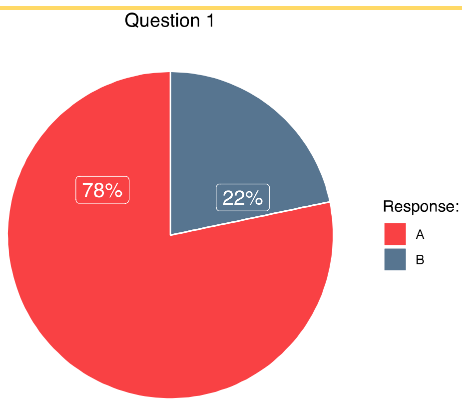
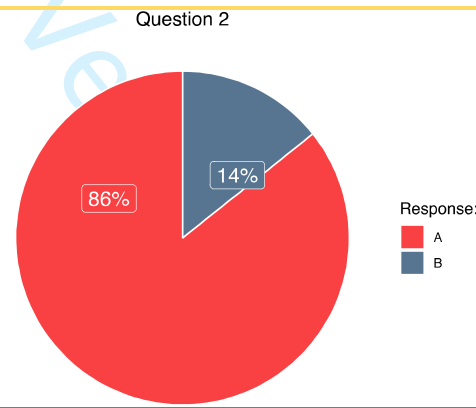



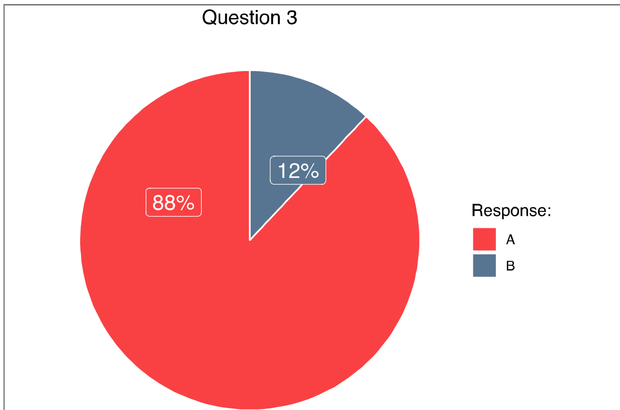



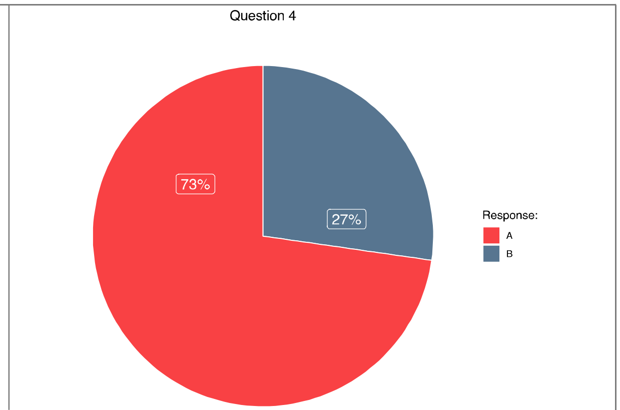




As mentioned in Section A.2, we asked each participant seven questions. However, not all participants responded to every question, resulting in some entries marked as "No Response (NR)." This is common in survey research. Accordingly, the reported percentages below are based only on the participants who provided a response to each question. For comprehensive results, Table A-1 presents the detailed responses to all questions.

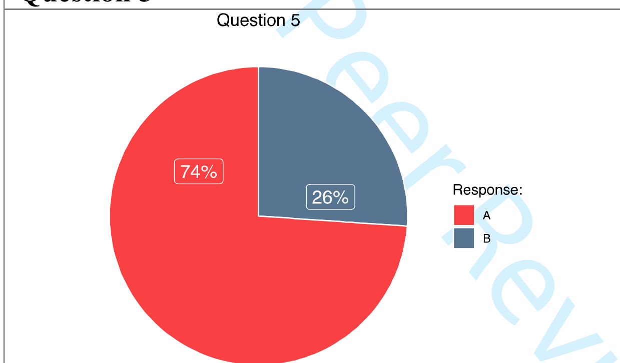
**Distribution of Responses**

| **Question 1** | **Question 2** |
|---|---|
|   |   |
| **Question 3** | **Question 4** |

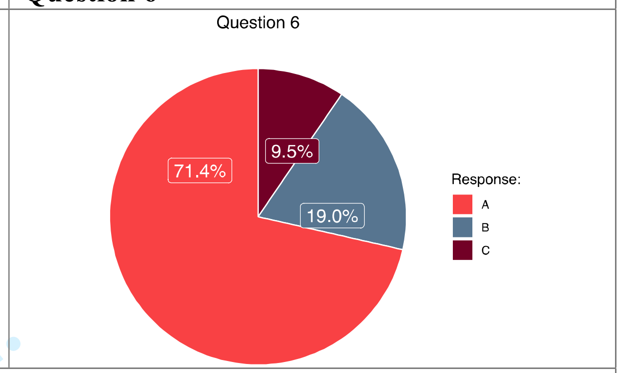



**Question 5**




**Question 6**




**Question 7**

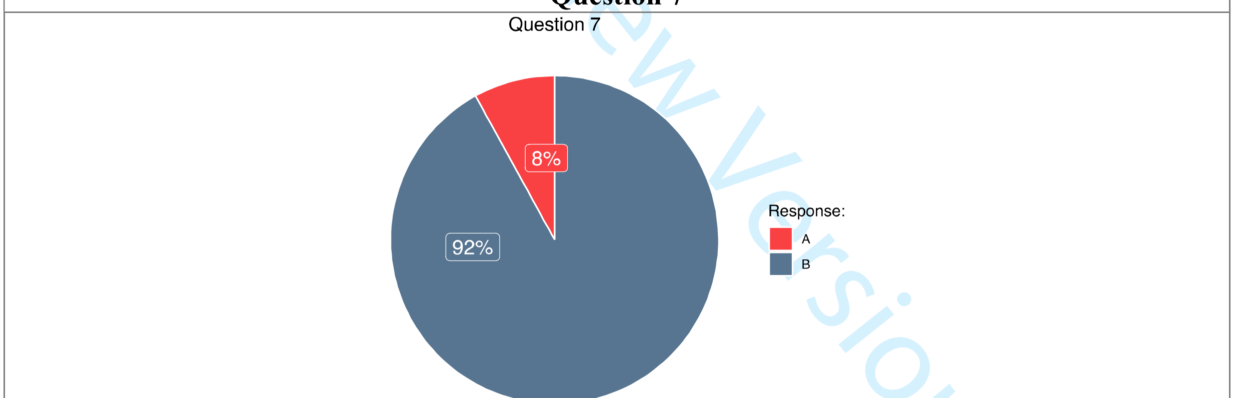





**Table A-1**

| Response Count (ALL) | Q1 | Q2 | Q3 | Q4 | Q5 | Q6 | Q7 |
|---|---|---|---|---|---|---|---|
| A | 18 | 18 | 22 | 16 | 17 | 15 | 2 |
| B | 5 | 3 | 3 | 6 | 6 | 4 | 23 |
| C | | | | | | 2 | |
| No Response (NR) | 2 | 4 | 0 | 3 | 2 | 4 | 0 |
| Total with NR | 25 | 25 | 25 | 25 | 25 | 25 | 25 |

## B. Summary of Notation

**Table B-1** Summary of notation

| Notation | Description |
|---|---|
| $a_t$ | decision variable |
| $\mathcal{A}_t$ | set of decisions |
| $\alpha^{\wp}_{jt}$ | probability that patient type $\wp$ faces terminal event $j$ in period $t$ |
| $b_t$ | measured BP |

| | |
|---|---|
| $\boldsymbol{B}$ | Bayesian learning strategy |
| $\boldsymbol{c_h}$ | a coefficient used in the formulation of $\alpha_{jt}^{\mathscr{p}}$ |
| $\boldsymbol{\delta_i}$ | decrements in QALYs as a result of medication $i$'s side-effects |
| $\boldsymbol{\Delta}$ | Degree of commitment to belief: threshold for defining surprise in the SIL strategy |
| $\boldsymbol{\Delta t}$ | time between decision epochs |
| $\boldsymbol{e}$ | error terms in the linear dynamic system in Eq. (1) |
| $\boldsymbol{\epsilon_i}$ | average medication $i$'s effectiveness (i.e., BP reduction) |
| $\boldsymbol{\eta}$ | average BP increase due to one year of aging |
| $\boldsymbol{\mathcal{H}}$ | value to go function |
| $\boldsymbol{h_t}$ | a component of the health state variable |
| $\boldsymbol{i}$ | medication index |
| $\boldsymbol{\mathcal{I}_t}$ | set of unprescribed medications in period $t$ |
| $\boldsymbol{\iota_t^h}$ | a coefficient used in the formulation of $\alpha_{jt}^{\mathscr{p}}$ |
| $\boldsymbol{j}$ | index for the terminal events |
| $\boldsymbol{\mathscr{j}}$ | index for the learning strategies |
| $\boldsymbol{k_j^{\mathscr{p}}}$ | a coefficient used in the formulation of $\alpha_{jt}^{\mathscr{p}}$ |
| $\boldsymbol{K_t}$ | Kalman gain |
| $\boldsymbol{\ell}$ | loss function for prediction errors |
| $\boldsymbol{\lambda}$ | discount factor |
| $\boldsymbol{LE_t^{\mathscr{p}}}$ | life expectancy |
| $\boldsymbol{m_t}$ | medication state |
| $\boldsymbol{M}$ | maximum number of medications on the list |
| $\boldsymbol{N}$ | maximum number of consecutive non-surprising office visits |
| $\boldsymbol{\mu_t^{\mathscr{j}}}$ | best estimate (or belief) regarding underlying BP in $t$ obtained through learning strategy $\mathscr{j}$ |
| $\boldsymbol{\mu_t'}$ | mean of BP regimen change |
| $\boldsymbol{n_t}$ | number of consecutive non-surprising visits |
| $\boldsymbol{o}$ | index for 24-hr BP monitoring |
| $\boldsymbol{\mathscr{p}}$ | patient type/characteristics |
| $\boldsymbol{P_t}$ | transition probability matrix |
| $\boldsymbol{\pi_t^{\mathscr{j}}(\theta_t)}$ | belief distribution obtained through learning strategy $\mathscr{j}$ |
| $\boldsymbol{\phi^2}$ | sum of SBPV and measurement noise |
| $\boldsymbol{Q_j}$ | one-period expected QALYs |
| $\boldsymbol{\bar{r}}$ | one-period expected reward function |
| $\boldsymbol{\hat{r}}$ | one-period average adjusted reward function |
| $\boldsymbol{R_{jt}^{\mathscr{p}}}$ | terminal reward in $t$ for patient type $\mathscr{p}$ who faces the terminal event $j$ |
| $\boldsymbol{\rho_t}$ | learning rate |
| $\boldsymbol{\varrho}$ | cumulative probability |
| $\boldsymbol{RV}$ | real BP variability |
| $\boldsymbol{\mathscr{s}_t}$ | surprise |
| $\boldsymbol{\sigma_{t,\mathscr{j}}^2}$ | variance term of the belief distribution |
| $\boldsymbol{\varsigma}$ | violation of assumption |
| $\boldsymbol{\sigma_b^2}$ | short-term BPV (SBPV) |
| $\boldsymbol{\sigma_\theta^2}$ | long-term BPV (LBPV) |
| $\boldsymbol{t}$ | decision epoch |
| $\boldsymbol{T}$ | decision horizon |
| $\boldsymbol{\tau^2}$ | device noise |
| $\boldsymbol{\theta_t}$ | true underlying BP mean |
| $\boldsymbol{v}$ | value function |
| $\boldsymbol{w_j}$ | QALYs of a patient who faces terminal event $E_j$ |
| $\boldsymbol{x_t}$ | state variable |
| $\boldsymbol{\hat{x}_t}$ | non-terminal health state |

| | |
|---|---|
| $\xi^{j}_{TF}$ | %share of the technology factor in the total value of information |
| $\xi^{j}_{HF}$ | %share of the human factor in the total value of information |
| $\bar{\xi}^{j}_{r}$ | average relative share of the technology factor vs human factor in total information value |
| $\boldsymbol{y}_t$ | health state |

## C. Transition Probability Formulation

The state transition function characterizes transitions to any state in the next period based on the state and decision in the current period. The transition function for medication states $\boldsymbol{m}_t$ is characterized by a simple deterministic function as it only updates $\boldsymbol{m}_{t+1}$ depending on whether a medication was added to $\boldsymbol{m}_t$. The transition for the non-medication states, i.e., $(h_t^{SIL}, n_t)$ is stochastic and governed by a Markov chain $P_t[h_{t+1}^{SIL}, n_{t+1}|h_t^{SIL}, n_t, a_t]$ with elements $p_t[h_{t+1}^{SIL}, n_{t+1}|h_t^{SIL}, n_t, a_t]$, which we characterize in two steps:

**1) Transitions among non-terminal health states**

We denote by $\tilde{P}_t[\mu_{t+1}^{SIL}, n_{t+1}|\mu_t^{SIL}, n_t, a_t]$, with elements $\tilde{p}_t[\mu_{t+1}^{SIL}, n_{t+1}|\mu_t^{SIL}, n_t, a_t]$, the transition probability for the non-terminal health states. The probability of such transition is equal to the probability of observing a BP measurement that induced the transition, i.e.,

$$\tilde{p}_t[\mu_{t+1}^{SIL}, n_{t+1}|\mu_t^{SIL}, n_t, a_t] = f_{\mathcal{B}_{t+1}}(b_{t+1}) \;\text{ IFF }\; \mu_t^{SIL}, n_t, a_t \xrightarrow{b_{t+1}} \mu_{t+1}^{SIL}, n_{t+1} \tag{A-1}$$

In other words, $b_{t+1}$ is the BP observation that has induced the transition and $f_{\mathcal{B}_{t+1}}$ characterizes the probability distribution of observing $b_{t+1}$.

**2) Transitions to the terminal health states**

The transitions in (A-1) are conditional on the absence of terminal events. To fully characterize the transition functions for all health states, we need three additional elements to account for the three terminal events in our model. Let, $\bar{\alpha}_{1t}(\mu_t^{SIL}, n_t, a_t)$, $\bar{\alpha}_{2t}(\mu_t^{SIL}, n_t, a_t)$, and $\bar{\alpha}_{3t}(\mu_t^{SIL}, n_t, a_t)$ denote the expected probabilities of transitioning from $(\mu_t^{SIL}, n_t)$, respectively, to the three terminal states, i.e., $E_1$, $E_2$, and $E_3$, under decision $a_t$. These parameters depend on the expectations over the true underlying BP, which is benchmarked with unbiased learning (i.e., KF). Taking the above considerations into account, the expected probability of facing $E_j$ can be formulated as:

$$\bar{\alpha}'_{jt}(\mu_t^{SIL}, n_t, a_t) = \left(1 - \bar{\alpha}_{3t}(\mu_t^{SIL}, n_t, a_t)\right)\bar{\alpha}_{jt}(\mu_t^{SIL}, n_t, a_t), j = \{1,2\} \tag{A-2}$$

We can now fully characterize the elements of the transition probability function $P_t$ as follows:

$$p_t[h_{t+1}^{SIL}, n_{t+1}|h_t^{SIL}, n_t, a_t] = \begin{cases} \left(1 - \sum_{j=1}^{3} \bar{\alpha}'_{3t}(\mu_t^{SIL}, n_t, a_t)\right)\tilde{p}_t[\mu_{t+1}^{SIL}, n_{t+1}|\mu_t^{SIL}, n_t, a_t]; \; h_t^{SIL}, h_{t+1}^{SIL} \in \mathcal{B}^{SIL} \\ \bar{\alpha}'_{jt}(\mu_t^{SIL}, n_t); \; h_t^{SIL} \in \mathcal{B}^{SIL}, h_{t+1}^{SIL} \in \mathcal{E} \\ 1; \; \{h_t^{SIL} \in \mathcal{E} \text{ or } n_t = N\}, h_{t+1}^{SIL} = E_3 \\ 0; \; \text{otherwise} \end{cases} \tag{A-3}$$

with $\bar{\alpha}'_{3t}(\mu_t^{SIL}, n_t, a_t) = \bar{\alpha}_{3t}(\mu_t^{SIL}, n_t, a_t)$.

## D. Conditions for the Structured Optimal Policy

In this section, we present the technical definitions and assumptions that are necessary to characterize optimal policy. For notational simplicity, we only focus on the health states $\hat{y}_t^{SIL} = (\mu_t^{SIL}, n_t)$ and characterize the structural properties in terms of $\mu_t^{SIL}$ and $n_t$. To save on the notations, we simply use $\mu_t$.

**Definition 1 [Increasing Failure Rate (IFR)]**: A transition probability matrix (and the associated Markov chain) is called IFR if its rows are in increasing stochastic order. This definition is equivalent to the well-known notion of first-order stochastic dominance (Chhatwal et al. 2010). IFR property has been commonly used in medical decision-making (Chhatwal et al. 2010) and has an intuitive interpretation in the context of this study. In summary, the IFRness of the transition probability matrix with $\mu_t$ means that the higher the BP level, the more likely is transitioning to higher BP levels or terminal events. The IFRness with $n_t$ implies that the higher the belief strength (which also implies a larger number of the consecutive non-surprising visits), the less likely becomes facing a surprise.

**Definition 2 [Superadditivity]**: A function $g(x, a)$ is superadditive if for $x^+ \geq x^-$ and $a^+ \geq a^-$, the following holds:

$$g(x^+, a^+) + g(x^-, a^-) \geq g(x^+, a^-) + g(x^-, a^+)$$

If reverse inequality holds, then $g(x, a)$ is said to be subadditive. An equivalent definition of a superadditive function is commonly stated as follows:

$$g(x^+, a^+) - g(x^-, a^+) \geq g(x^+, a^-) - g(x^-, a^-)$$

Note that, in our case, $a_t^-$ (less aggressive action) and $a_t^+$ (more aggressive action) indicate doing nothing and prescribing a new medication, respectively, i.e., $a_t^- = 0$ and $a_t^+ = i \in \mathcal{A}_t$. This condition implies that a superadditive (resp. subadditive) function has non-decreasing (resp., nonincreasing) differences in $x$ with respect to $a$.

Based on the above definitions, we now introduce the following technical assumptions that would be needed for establishing the structural properties of the optimal policies for **MDP**$^{SIL}$. In all the statements, if a variable is not indicated explicitly, it implies that it is kept constant.

**ASSUMPTION 1**. *$r_t(\mu_t, n_t, a_t)$ and $r_T(\mu_t, n_t)$ are:*

*(i)* *nonincreasing in $\mu_t$,*

*(ii)* *nondecreasing in $n_t$.*

**ASSUMPTION 2**. *The transition probability matrix $P_t$ is IFR in (i) $\mu_t$ and (ii) $n_t$.*

**ASSUMPTION 3**. *$r_t(\mu_t, n_t, a_t)$ is:*

(i) *superadditive in* $\mu_t$ *and* $a_t$,

(ii) *superadditive in* $n_t$ *and* $a_t$.

**ASSUMPTION 4**. *The transition matrix* $P_t$ *is such that:*

(i) $q_{1t}(k_{1t}, \mu_t, n_t, a_t) = \sum_{k_{1t}=\mu_t'} p_t(k_{1t}, n_{t+1}|\mu_t, n_t, a_t)$ *is subadditive in* $\mu_t$ *and* $a_t$,

(ii) $q_{2t}(k_{2t}, \mu_t, n_t, a_t) = \sum_{k_{2t}=n_t'} p_t(\mu_{t+1}, k_{2t}|\mu_t, n_t, a_t)$ *is superadditive in* $n_t$ *and* $a_t$.

Although similar assumptions are commonly employed in the literature to characterize the structural properties of the optimal policies (Puterman 2005), in our context, the above assumptions yield intuitive interpretations as follows. **Assumption 1(i)** means that the expected quality of life and life expectancy do not increase with BP. **Assumption 1(ii)** implies that the expectations for the quality of life and life expectancy do not decrease as the belief strength increases. First, according to the Framingham risk equations, the increased CVD risks as a result of a one mmHg increase in BP is larger than the decreased CVD risk as a result of a one mmHg decrease in BP (D'Agostino et al. 2008). Second, increasing $n_t$ implies a narrower normal distribution, hence a lower probability of reaching higher levels of BP. Therefore, our expectation for facing CVD events decreases with $n_t$, leading to higher expectations for the quality of life and life expectancy. We previously discussed the intuitive implications of **Assumption 2** regarding the IFR property of the transition probability matrix. **Assumption 3(i)** implies that the benefit of medication prescription does not decrease with increased BP. This is also intuitive because, at higher BP levels, the CVD risks outweigh the medication side-effects. Similarly, **Assumption 3(ii)** implies that the benefit of medication prescription does not decrease with increased belief strength. **Assumption 4(i)** means that the difference between doing nothing versus prescribing medication in terms of the probability of progressing to a certain BP level or higher, including terminal events, does not decrease with the BP level. This follows because of the greater importance of BP lowering at higher BP levels. Finally, **Assumption 4(ii)** implies that when prescribing medications, the difference between the probability of facing no-surprise under $n_t^+$ (i.e., higher confidence) versus $n_t^-$ (i.e., lower confidence) does not decrease. In other words, medication prescription does not lead to a decrease in the belief strength.

Due to the page limitations, the proofs of the Theorems 1–2 are provided in the Technical Notes document (for review only). In this document, we also verify whether the calibrated model parameters in Appendix F satisfy Assumptions 1–4 and estimate a bound on their maximum violation. Since these violations are small, the optimal policy can be characterized by a threshold level for each medication.

## E. BP Uncertainty Scenarios

In our study, we have taken three levels, denoted by $L$ (i.e., low), $M$ (i.e., medium), and $H$ (i.e., high), for each of the BP uncertainty parameters, namely, $\sigma_\theta$, $\sigma_b$, and $\tau$. This results in a total of 27 scenarios, according to Table E-1. Note that the general magnitude of variability increases from Scenario 1 to Scenario 27

**Table E-1.** BP Uncertainty Scenarios

| Region | LBPV ($\sigma_\theta$) | SBPV ($\sigma_b$) | Measurement Noise ($\tau$) | Scenarios |
|---|---|---|---|---|
| **1** | L | L, M, H | L, M, H | 1–9 |
| **2** | M | L, M, H | L, M, H | 10–18 |
| **3** | H | L, M, H | L, M, H | 19–27 |

## F. Parameter Estimation

The primary source of data for our study consists of two real-life datasets collected by tracking 376 patients over six years at the Vascular Health Unit in Montreal General Hospital, Canada. As the secondary source of data, we consulted the medical literature. Our two real-life datasets were collected through two carefully designed BP measurement settings. In the first setting, at each office visit, a rigorous protocol was followed to measure participants' BP using Automated Office Blood Pressure (AOBP) technology, which is described in the main paper. We use this setting for measuring a good proxy for the true underlying BP (i.e., the ground truth) $\theta_t$ and its long-term (i.e., visit-to-visit) variability $\sigma_\theta^2$. This type of setting is commonly used in clinical studies to measure long-term BP variability (Mehlum et al. 2018, Parati et al. 2013). In the second BP measurement setting, the patients were wearing a device for 24–48 hours to have their BP measured every 20–30 minutes. This setting, which is also called 24-hour or Ambulatory BP Monitory (ABPM), allows us to estimate the short-term BP variability as well as the measurement noise around the true underlying BP captured in the first BP measurement setting. In clinical studies, this BP measurement setting is commonly used to measure short-term BP variability (Mehlum et al. 2018, Parati et al. 2013). Since the device includes noise, this setting also allows us to estimate the measurement noise. In our study, $\phi^2$ denotes the sum of the two variance terms (please see Eq. (1) in the main manuscript). Because the measurements in the first dataset were made during the day, when the two datasets had to be linked, we filtered the second dataset to the daytime records of the patients who also participated in the first BP measurement setting within one week. Table F-1 provides the details of the baseline characteristics of the patients in our datasets. Finally, we used the cubic spline technique to interpolate the missing office BP observations (Alagoz 2004).

**Table F-1** Baseline participant characteristics in the two BP measurement settings

| Data Feature | Value |
|---|---|
| total number of cases (i.e., patients) | 376 |
| male | 199 (53%) |
| min age | 21 |
| max. age | 84 |
| min. number of visits | 1 |
| max. number of visits | 8 |
| number of visits mean | 3.6 |
| number of visits median | 3 |
| total number of visits | 805 |
| earliest visit | Aug. 2008 |
| latest visit | Apr. 2014 |
| total number of valid cases in the 1st dataset (i.e., patients with more than one visit) | 218 (58%) |

| | |
|---|---|
| total number of valid cases in the 2nd dataset (i.e., patients who participated in both measurement settings and their participations are not more than one week away) | 183 |
| total number of measurements in the 2nd datasets | 5,955 (daytime=2,977) |
| total number of valid measurements for male patients | 3,692 (62%) |

In what follows, we provide further details regarding the parameter estimation in our study.

**Long-Term BP Variability ($\sigma_\theta$).** We estimated this parameter using the first dataset. To obtain the patient-specific estimations indexed by $\wp$, we stratified the data on the patient characteristics. In our study, our data allowed us to define $\wp = \{\text{male, female}\}$. With larger datasets, where further stratifications are possible, $\wp$ can be expanded to include more characteristics. For each stratification, we estimated $\sigma_\theta$ as the standard deviation of the BP real variability (RV) defined as follows:

$$RV_t = \theta_{t+1} - \theta_t \tag{A-4}$$

RV has been extensively used in the medical literature as one of the most reliable measures of long-term BP variability $\sigma_\theta$ (Mehlum et al. 2018). To calculate $\sigma_\theta^M$, $\sigma_\theta^L$ and $\sigma_\theta^H$, we used data from the patients with at least two observations. Following the procedure in Mehlum et al. (2018), we reported the lowest and highest quintiles of the long-term BP variability as $\sigma_\theta^L$ and $\sigma_\theta^H$, respectively. The estimation results are as follows:

**Table F-2** Estimated values of $\sigma_\theta$

| | $\boldsymbol{\sigma_\theta^M}$ | **95% CI** | $\boldsymbol{\sigma_\theta^L}$ | $\boldsymbol{\sigma_\theta^H}$ |
|---|---|---|---|---|
| **male** | 12.35 | [11.59, 13.22] | 5.93 | 26.06 |
| **female** | 10.45 | [9.74, 11.27] | 4.28 | 18.59 |

These parameters suggest that male patients experience significantly larger LBPV than female patients.

**Short-term BP variability and noise ($\phi$).** We estimated this parameter using both datasets. In linking the two datasets, we matched them using only the patients who participated in both measurement settings within a maximum of one week. In doing this, we only used the daytime BP records in the 2nd dataset. Next, we computed $\phi$ for each clinic visit and its associated daytime BP measurements. Finally, $\phi^M$ was estimated as the mean of all these $\phi$'s, while $\phi^L$ and $\phi^H$ corresponded to the lowest and highest quartiles of the $\phi$'s. We assumed an equal share of $\tau^2$ and $\sigma_b^2$ in the corresponding $\phi^2$. Table F-3 summarizes the estimation results.

**Table F-3** Estimated values of $\phi$

| | $\boldsymbol{\phi^M}$ | **95% CI** | $\boldsymbol{\phi^L}$ | $\boldsymbol{\phi^H}$ |
|---|---|---|---|---|
| **male** | 9.13 | [5.95,12.31] | 4.15 | 16.25 |
| **female** | 7.79 | [5.67,12.31] | 3.54 | 15.66 |

These estimations suggest that when using the ABPM devices, male patients exhibit a larger fluctuation in their daily BP readings than female patients (considering all components of $\phi$).

**Optimal Surprise Threshold ($\Delta^*$).** To estimate this parameter, which pertains only to the SIL strategy, we minimized an expected loss function $\ell(\Delta)$ defined as the mean squared errors of predicting the true underlying BP through the $SIL(\Delta)$ strategy. In other words:

$$\Delta^* = \underset{\Delta}{\operatorname{argmin}}\ \mathbb{E}[\ell_t(\Delta, \wp)] \tag{A-5}$$

where:

$$\ell_t(\Delta, \wp) = \left(\theta_t - \mu_t^{SIL}(\Delta, \wp)\right)^2 \tag{A-6}$$

We used the gradient descent algorithm to search for $\Delta^*$, and when multiple solutions existed, we considered the smallest one. Note that a similar loss function minimization procedure is used in a recursive way to estimate the Kalman gain parameter in the KF strategy. Due to our definition of surprise, we could not find a closed-form expression for $\Delta^*$. Therefore, we estimated $\Delta^*$ numerically using data in the following way. First, we chose the patients with at least five office visits from the 1st dataset along with their underlying BP mean $\theta_t$. This resulted in a dataset of 94 patients. Next, for each patient, we matched their office visits with the daytime BP readings from the 2nd data (which were at most one week apart). In the case of missing daytime observations corresponding to an office visit, we generated 24 random BP values from the LDS Eq. (1) in the paper using $\theta_t$ and $\phi^M(\wp)$. Each of these BP readings denoted as $b_o$, provides us with an estimation for $\mu_{t,o}^{SIL}(\Delta, \wp)$. Next, we estimated $\mu_t^{SIL}(\Delta, \wp)$ as the mean of all $\mu_{t,o}^{SIL}(\Delta, \wp)$. Next, equation (A-6) gives us the prediction loss for the patient's visit $t$. Finally, we averaged the results for all patients and visits. The results of our estimations reveal that $\Delta^* = 9.9\ mmHg$.

We compared the $SIL(\Delta^*)$ and KF strategies based on their prediction errors measured through mean absolute error (MAE), root mean square error (RMSE), and the mean absolute percentage error (MAPE), defined as:

$$MAE = \mathbb{E}\left[\left|\theta_t - \mu_t^{SIL}(\Delta, \wp)\right|\right] \tag{A-7}$$

$$RMSE = \sqrt{\mathbb{E}\left[\left(\theta_t - \mu_t^{SIL}(\Delta, \wp)\right)^2\right]} \tag{A-8}$$

$$MAPE = \mathbb{E}\left[\left|\frac{\theta_t - \mu_t^{SIL}(\Delta, \wp)}{\theta_t}\right|\right] \tag{A-9}$$

Table F-4 illustrates the results of our comparisons.

**Table F-4** Prediction Errors in $SIL(\Delta^*)$ and $KF$ from our real-life data

| | $\boldsymbol{SIL(\Delta^*)}$ | $\boldsymbol{KF}$ |
|---|---|---|
| **MAE (mmHg)** | 4.43 | 3.27 |
| **RMSE (mmHg)** | 5.33 | 3.99 |
| **MAPE (%)** | 3.43 | 2.54 |

These results suggest that the magnitude of error in both learning strategies is small and $KF$ performs only slightly better than $SIL(\Delta^*)$. For all other BP variability scenarios (i.e., the combinations of $\phi$ and $\sigma_\theta$) for which we did not have enough real data, we used bootstrapping to create simulated datasets based on the LDS Eq. (1) and the estimated parameters above. For each bootstrapped data, we simulated 100 patients (maintaining the gender proportion in the data) with 20 office BP measurements, each associated with 24 ambulatory daytime BP observations, preserving their distributional dependencies. Finally, for each dataset, we computed $\Delta^*$ as well as the prediction errors for the $SIL(\Delta^*)$ and $KF$ strategies, as explained above. For each scenario, we created 10 simulated datasets. Table F-5 provides a summary of the average results for each scenario, and Figure F-1 illustrates the trend of errors.

**Table F-5** Prediction errors in $SIL(\Delta^*)$ and $KF$ from simulated datasets under different BP variability scenarios $(\phi, \sigma_\theta)$

| | | | $SIL(\Delta^*)$ | | | $KF$ | | |
|---|---|---|---|---|---|---|---|---|
| **No** | **Data Scenario** | $\Delta^*$ | **MAE** | **RMSE** | **MAPE** | **MAE** | **RMSE** | **MAPE** |
| **1** | $\phi^L, \sigma_\theta^L$ | 4.5 | 4.02 | 4.86 | 3.13 | 2.89 | 3.50 | 2.24 |
| **2** | $\phi^L, \sigma_\theta^M$ | 6.5 | 3.09 | 3.72 | 2.38 | 2.36 | 2.86 | 1.82 |
| **3** | $\phi^L, \sigma_\theta^H$ | 10.0 | 2.29 | 2.80 | 1.77 | 2.00 | 2.43 | 1.54 |
| **4** | $\phi^M, \sigma_\theta^L$ | 8.3 | 5.04 | 6.20 | 3.90 | 3.41 | 4.20 | 2.65 |
| **5** | $\phi^M, \sigma_\theta^H$ | 14.2 | 3.56 | 4.35 | 2.76 | 2.81 | 3.41 | 2.18 |
| **6** | $\phi^H, \sigma_\theta^L$ | 14.8 | 6.44 | 7.88 | 4.99 | 4.41 | 5.31 | 3.41 |
| **7** | $\phi^H, \sigma_\theta^M$ | 14.4 | 5.54 | 6.81 | 4.31 | 3.87 | 4.73 | 3.01 |
| **8** | $\phi^H, \sigma_\theta^H$ | 18.6 | 4.88 | 5.95 | 3.78 | 3.66 | 4.44 | 2.84 |

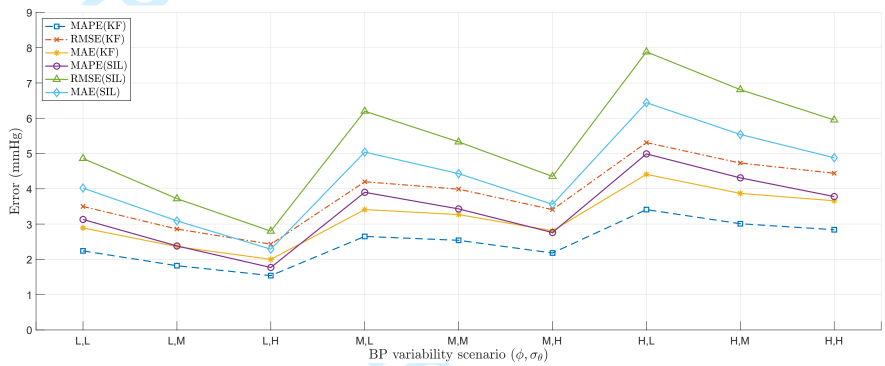



**Figure F-1** Trend of prediction errors for SIL and KF under different BP variability scenarios $(\phi, \sigma_\theta)$

These results suggest that the prediction errors for both learning strategies are relatively small, and $KF$ outperforms $SIL(\Delta^*)$ with small margins. Also, the prediction errors for both learning strategies change similarly with the BP variability scenarios. Finally, $\Delta^*$ increases with the BP variability scenario and the change is more sensitive to $\phi$, indicating that the observed change in BP readings becomes less reliable when BP variability, particularly the one caused by the short-term BP or measurement noise, increases. For the BP uncertainty scenarios not listed above, we interpolated $\Delta^*$ from the closest two scenarios.

**Medication Effectiveness ($\epsilon_i$).** We used the estimations provided through the seminal paper by Law et al. (2009) for $\epsilon_i$ for each of the five classes of medications used in our study. Following their procedure, we estimated $\epsilon_i$, under the standard medication dose, along with the $R^2$ for the fitted regression, as follows:

$$\epsilon_1 = \epsilon_{ACEI}(SBP) = 0.0347SBP; \;\; R^2 = 0.9651$$
$$\epsilon_2 = \epsilon_{\mathrm{DU}}(SBP) = 0.0360SBP; \;\; R^2 = 0.9477$$
$$\epsilon_3 = \epsilon_{CCB}(SBP) = 0.0364SBP; \;\; R^2 = 0.9380$$
$$\epsilon_4 = \epsilon_{BB}(SBP) = 0.0373SBP; \;\; R^2 = 0.9106$$
$$\epsilon_5 = \epsilon_{ARB}(SBP) = 0.0424SBP; \;\; R^2 = 0.4155$$

These parameters suggest that among the five classes of antihypertensives, ACEI and ARB are the least and most effective medications for BP reductions, respectively.

**Probability of Nonfatal Events ($\alpha_{1t}$ and $\alpha_{2t}$):** We used the estimations of $\alpha_{1t}$ and $\alpha_{2t}$ from the Framingham Study in D'Agostino et al. (2008), one of the most popular epidemiological studies conducted to date. The study provides simple gender-specific equations for calculating the risk of CVD as a function of age ($t$), total cholesterol ($CLT$), High-Density Lipoprotein cholesterol ($HDL$), Systolic Blood Pressure ($SBP$), Smoking status ($SMK$) and Diabetes status ($DIAB$). The study also provides calibration parameters for quantifying the risks of CHD or ST based on the computed general CVD risk. We used the following set of equations to estimate $\alpha_{jh}$ for CHD ($j = 1$) and ST ($j = 2$) for male ($\wp = 1$) and female ($\wp = 2$):

$$\alpha_{jt}^{\wp}(t, CLT, HDL, SBP, SMK, DIAB) = k_j^{\wp}\left(1 - c_{\wp}{}^{\iota_t^{\wp}(t,CLT,HDL,SBP,SMK,DIAB)}\right) \qquad \text{(A-10)}$$

where $k_j^{\wp}$ is the gender-specific calibration parameter for the two nonfatal events. The elements of the above risk function have been estimated as follows:

| $k_j^{\wp}$ | $j = 1$ | $j = 2$ |
|---|---|---|
| $\wp = 1$ | 0.7174 | 0.1590 |
| $\wp = 2$ | 0.6086 | 0.2385 |

| | $c_{\wp}$ |
|---|---|
| $\wp = 1$ | 0.88936 |
| $\wp = 2$ | 0.95012 |

| | $\iota_t^{\wp}(t, CLT, HDL, SBP, SMK, DIAB)$ | |
|---|---|---|
| $\wp = 1$ | $exp\left\{\begin{matrix} 3.06117Ln(t) + 1.12370Ln(CLT) - 0.93263Ln(HDL) \\ +1.93303Ln(SBP) + 0.65451SMK + 0.57367DIAB - 23.9802 \end{matrix}\right\}$ | (A-11) |
| $\wp = 2$ | $exp\left\{\begin{matrix} 2.3288Ln(t) + 1.20904Ln(CLT) - 0.70833Ln(HDL) \\ +2.67157Ln(SBP) + 0.52873SMK + 0.69154DIAB - 26.1931 \end{matrix}\right\}$ | (A-12) |

The statistical details regarding the above estimations can be found in D'Agostino et al. (2008).

**Probability of Non-CVD-Mortality ($\alpha_{3t}$):** We used Lewington et al. (2002), another seminal study in the medical literature, to estimate the risk of non-CVD-mortality. The fitted curve estimates the probability of non-CVD mortality as an exponential function of age ($t$) as follows:

$$\alpha_{3t}(t) = 5 \times 10^{-5} e^{0.0824t}; \;\; R^2 = 0.9988 \qquad \text{(A-13)}$$

**Post-Events Life Expectancy ($R_{1t}$ and $R_{2t}$):** We defined the Lump-Sum-Terminal Rewards for the non-fatal events, namely $R_{1t}$ and $R_{2t}$, as the QALE accrued after facing CHD and ST, respectively. These elements quantify both the quantity and quality of years of life after facing the two nonfatal events. A clinical study by Hannerz and Nielsen (2001) provides the estimations for the gender-specific post-CVD Life Expectancy (LE), which can be summarized as follows:

$$R_{jt}^{\wp} = \omega_j \times LE_t^{\wp} \qquad \wp \in \{1,2\}, j \in \{1,2\} \qquad \text{(A-14)}$$

where $\omega_1$ and $\omega_2$ are the QALYs under CHD and ST, respectively, and $LE_t^{\wp}$ is estimated for male ($\wp = 1$) and female ($\wp = 2$) using the following equations:

| | $LE_t^{\wp}$ | |
|---|---|---|
| $\wp = 1$ | $0.0073t^2 - 1.3673t + 68.683; \;\; R^2 = 0.9994$ | (A-15) |
| $\wp = 2$ | $0.0073t^2 - 1.4696t + 76.425; \;\; R^2 = 0.9999$ | (A-16) |

The above equations suggest that female patients have a longer post-CVD life expectancy.

**Other parameters:** We summarized the estimated values for all other parameters in Table F-6.

**Table F-6** Estimated Values of the Remaining Parameters

| Parameter | Estimated Value | Source |
|---|---|---|
| Gender-specific slope of the linear blood pressure trend over time ($\boldsymbol{\eta}$) | $\eta_M$=0.30 | (Wilkins et al. 2010) |
| | $\eta_F$= 0.37 | |
| QALY under nonfatal events ($\boldsymbol{\omega_j}$) | $\omega_{CHD}$=0.77 | (Lovibond et al. 2011) |
| | $\omega_{ST}$=0.63 | |
| Discount Factor ($\boldsymbol{\lambda}$) | 0.996 | (Lovibond et al. 2011) |
| Medication Side-Effects ($\boldsymbol{\delta_i}$) | $\delta_1$=0.0048 | (Lawrence et al. 1996, Mason et al. 2014, Tengs and Wallace 2000) |
| | $\delta_2$=0.0049 | |
| | $\delta_3$=0.0050 | |
| | $\delta_4$=0.0051 | |
| | $\delta_5$=0.0058 | |

We could not find any reliable data or study for a reasonable estimate of $\delta_i$, i.e., the decrement in the quality of life because of medication side-effects. Therefore, we prorated these parameters so that (a) they provide the same medication ordering as in Law et al. (2009) and Lawrence et al. (1996), (b) they are proportional to their effectiveness as in Law et al. (2009), while (c) they maintain a reasonable scale for the antihypertensive medication side-effects as in Tengs and Wallace (2000).

## G. Further Analyses of DQD

### G-1- Decision Quality Decrement with Respect to Decision Flexibility

We studied how decision quality decrement changes with the level of decision flexibility regarding the number of medications on the physician's list. To this end, we vary the total number of medications on the physician's list from $M = 1$ to $M = 5$. Fig G1 shows that decision quality decrement increases with decision flexibility. This is intuitive because, with increased flexibility, the optimal prescription decision (i.e., if, which, and when a medication should be prescribed) becomes more sensitive to the underlying BP regimen. Hence, knowing the patient's true BP level becomes more valuable as the number of potential medications increases. Second, DQD increases more rapidly under both $\Delta^L$ and $\Delta^H$. This implies that judgment bias in either direction would result in increasingly lower payoffs as the number of medications increases. Finally, similar to our previous observations, decision quality decrement is the highest under the under-estimation bias (i.e., $SIL(\Delta_H)$). In other words, the new technology is most valuable when the full list of medications (i.e., highest decision flexibility) is available to a physician who is overcommitted to their beliefs.

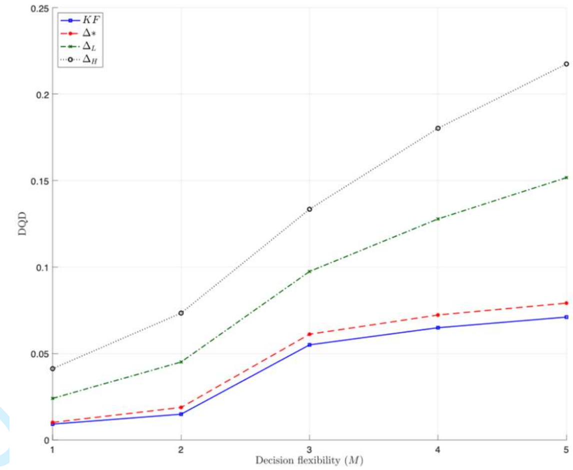



**Fig G1** Changes in the value of information with respect to decision flexibility

## G-2- The Relative Contribution of Technology Factor vs. Human Factor to DQD

To further compare the role of the technology versus human factor, we define $\xi_{TF}^{\acute{\jmath}}$ and $\xi_{HF}^{\acute{\jmath}}$ as the shares (i.e., %contributions) of the two factors in the total DQD, respectively. In other words:

$$\begin{aligned} \xi_{TF}^{\acute{\jmath}} &= DQD_{TF}^{\acute{\jmath}}/DQD^{\acute{\jmath}} = (v^{PI} - v^{KF})/(v^{PI} - v^{\acute{\jmath}}) \\ \xi_{HF}^{\acute{\jmath}} &= DQD_{HF}^{\acute{\jmath}}/DQD^{\acute{\jmath}} = (v^{KF} - v^{\acute{\jmath}})/(v^{PI} - v^{\acute{\jmath}}) \end{aligned} \tag{A-17}$$

Therefore, $\xi_{TF}^{\acute{\jmath}}$ and $\xi_{HF}^{\acute{\jmath}}$ decompose the contributions of the technology factor and human factors to the total DQD. In other words, $\xi_{TF}^{\acute{\jmath}} = 1$, if DQD is entirely due to the technology factor. Fig G2 summarizes the average relative share of the technology factor versus that of the human factor (i.e., the mean of $\xi_{HF}^{\acute{\jmath}}/\xi_{TF}^{\acute{\jmath}}$ denoted by $\bar{\xi}_r^{\acute{\jmath}}$) across the 27 scenarios under the three Δ values. Accordingly, under $SIL(\Delta^*)$, on average, the human factor has a significantly lower contribution than the technology factor ($\bar{\xi}_r^{SIL(\Delta^*)} = 0.14;\ 95\%\ CI = [0.10, 0.19]$), whereas under $SIL(\Delta_L)$, the two factors are equally important for $SIL(\Delta_L)$, ($\bar{\xi}_r^{SIL(\Delta_L)} = 0.98;\ 95\%\ CI = [0.92, 1.03]$), and under $SIL(\Delta_H)$, the human factor is around 73% more important than the technology factor ($\bar{\xi}_r^{SIL(\Delta_H)} = 1.73;\ 95\%\ CI = [1.37, 2.09]$).

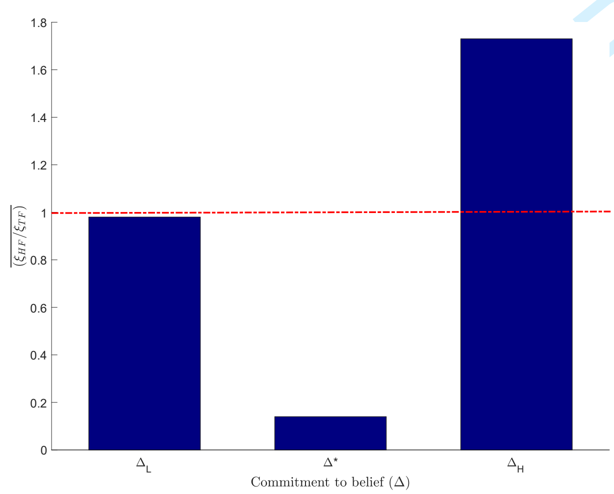



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**Figure G-2** Average share of human factor relative to the technology factor in DQD

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# TECHNICAL NOTES

# On The Interplay Between Clinical Inertia and Blood Pressure Measurement Technology in Hypertension Management

In this document, we provide the technical details regarding the structural properties of the optimal solutions (i.e., threshold-type policy) and the mathematical proofs. Please see Appendix D for the technical definitions and assumptions used for characterizing the optimal policy. In section 1, we provide the main results regarding the structured characterization of the optimal policy under $\mathbf{MDP}^{SIL}$. The proofs are provided in section 2, where we also provide Theorem 2 that states $\mu_t^*(n_t)$ is nonincreasing with $n_t$ and $t$. Finally, to verify whether the calibrated model parameters in our study (please see Appendix F) satisfy Assumptions 1–4 in Appendix D, we estimate a bound on their maximum violation in Section 3. Since these violations are small, the optimal policy can be characterized by a threshold level for each medication (for an illustration of the threshold policy, please see Section 5.3).

## 1. Structural Properties of the Optimal Solutions

In this section, we establish that there exist a minimum BP level and a belief strength level for prescribing each medication. While such a threshold policy is optimal for all three MDPs discussed above, we will focus on $SIL$, because its state space is more involved than the other two MDPs. The typical strategy employed to characterize the optimality of threshold policy includes a proof that the value function is monotone in the states and the action-value function has monotone differences in state-action pairs, a proof that these properties are preserved under the MDP operator, and an induction-based argument to show that the desired structural properties hold for all decision epochs (Puterman 2005). We need to adapt the approach for the $SIL$ strategy to account for the additional state $n_t$ as opposed to unidimensional state space for the optimization models under $PI$ and $KF$.

**Theorem 1.** *Under Assumptions 1–4, the following hold true:*

1. *In each period $t$, medication state $\mathbf{m}_t$, and belief strength $n_t$, there exists a BP threshold $\mu_t^*$ such that:*

$$a_t^*(\hat{x}_t^{SIL}) = \begin{cases} 0, & \mu_t < \mu_t^*(n_t) \\ i, & \mu_t \geq \mu_t^*(n_t) \end{cases} \tag{1}$$

2. *In each period $t$, medication state $\mathbf{m}_t$, and BP state $\mu_t$, there exists a belief strength threshold $n_t^*$ such that:*

$$a_t^*(\hat{x}_t^{SIL}) = \begin{cases} 0, & n_t < n_t^*(\mu_t) \\ i, & n_t \geq n_t^*(\mu_t) \end{cases} \tag{2}$$

Theorem 1 indicates that there is a minimum threshold for the estimated BP level and its corresponding belief strength in making decisions about prescriptions. According to these Theorems (not shown for $a_t^*$ with $t$), the thresholds exhibit pairwise inverse relationships between $\mu_t$, $n_t$, and $t$, which means that as one state increases, the threshold for another state decreases, suggesting important clinical implications. First, the threshold policies concerning $\mu_t$ and $n_t$ suggest that the lower the belief strength, the higher the minimum level of BP state to prescribe medication and vice-versa. From the perspective of clinical inertia, these structural properties provide actionable insights into the impact of BP uncertainty on the optimal prescription thresholds. This is because clinical inertia, characterized by delayed treatment initiation or adjustment, often stems from uncertainty and the resulting low confidence in BP status. The thresholds reveal that when confidence is low (low n), higher BP levels (higher $\mu$) are required to prescribe medications. This delay avoids unnecessary interventions unless the BP is excessively high. Conversely, increased confidence (high *n*) enables physicians to act decisively at lower BP levels, mitigating inertia and promoting timely interventions. This mitigates the risk of delayed treatment by facilitating timely interventions in line with evidence. While inertia can lead to suboptimal outcomes due to delayed action, it also serves a protective role by preventing premature decisions based on noisy or unreliable measurements. These findings underscore the importance of enhancing physician confidence through accurate measurements and robust learning mechanisms. While delay (i.e., inertia) is often viewed negatively, our analysis demonstrates that not all delays are suboptimal; however, the key question is when a delay represents the optimal decision and when it does not. Our decision analytics framework makes such distinctions, providing clinically meaningful tools and insights. The derived thresholds indicate that delay can be an optimal strategy when additional information is required to mitigate uncertainty, thereby enabling evidence-based interventions. Conversely, delays become suboptimal when they prolong high BP conditions unnecessarily, compromising patient outcomes. These findings advance our understanding of clinical inertia, offering actionable strategies for balancing caution with timely action.

The 2018 ESC/ESH HTN guideline by the European Heart Association has highlighted the inverse relationship between BP severity (equivalent to our $\mu_t^*$) and the number of office visits (equivalent to our $n_t^*$) (Williams et al. 2018). Similarly, the higher the BP level, the lower the minimum belief strength needed for prescribing a medication, and vice versa. Second, the BP threshold for medication prescription does not increase with age. Although controversial in the medical community (Myers et al. 2015), this observation is more consistent with important studies, such as the well-known SPRINT study (SPRINT Research Group 2015) and an empirical study based on the observations of 8.8 million patients in the US (Fletcher et al. 2017). Both studies indicate that older adults should maintain a lower BP than younger adults. Moreover, in a recent 5-year longitudinal study of 3,627 community-dwelling individuals aged 65 years or older, the presence of an elevated BP was associated with a significant increase in the CVD risks (Myers et al. 2015). Jaeger et al. (2022) argue that the benefit from the HTN treatment increases with age as aging increases CVD risk.

## 2. Proofs

While the proofs are provided for **MDP**$^{\mathbf{SIL}}$ with state $x_t^{SIL}$, a similar procedure can be followed for **MDP**$^{\mathbf{KF}}$ with state $x_t^{KF}$ and **MDP**$^{\mathbf{PI}}$ with state $x_t^{PI}$. For notational convenience, we drop superscript $SIL$ from all notations.

To characterize the structure of the optimal policies as in Theorem 1, first, we show a set of preliminary results pertaining to the value function. Note that the value function satisfies the Bellman equation provided in Eq. 16 in the main manuscript, and can be written as follows:

$$v_t(\mu_t, n_t, \boldsymbol{m}_t) = \max_{a_t \epsilon \mathcal{A}_t} [\mathcal{H}_t(\mu_t, n_t, \boldsymbol{m}_t, a_t)] \tag{3}$$

where $\mathcal{H}_t(\mu_t, n_t, \boldsymbol{m}_t, a_t)$ is the value-to-go function at time $t$ when the prescription decision is set to $a_t$ for a patient whose non-terminal health state is $\mu_t$, which has been obtained based on $n_t$ consecutive non-surprising observations while the his/her current list of prescribed medications is $\boldsymbol{m}_t$. In other words:

$$\mathcal{H}_t(\mu_t, n_t, \boldsymbol{m}_t, a_t) = \bar{r}_t(\mu_t, n_t, \boldsymbol{m}_t, a_t) + \lambda^{\Delta t} \sum_{\mu_{t+1}} p_t(\mu_{t+1}, \cdot \mid \mu_t, n_t, a_t) v_{t+1}(\mu_{t+1}, \cdot, \boldsymbol{m}_{t+1}) \tag{4}$$

where

$$\begin{aligned} p_t(\mu_{t+1}, \cdot \mid \mu_t, n_t, a_t) v_{t+1}(\mu_{t+1}, \cdot, \boldsymbol{m}_{t+1}) &= \sum_{n_{t+1}=0}^{n_t+1} p_t(\mu_{t+1}, n_{t+1} | \mu_t, n_t, a_t) v_{t+1}(\mu_{t+1}, n_{t+1}, \boldsymbol{m}_{t+1}) \\ = p_t(\mu_{t+1}, 0 | \mu_t, n_t, a_t) v_{t+1}(\mu_{t+1}, 0, \boldsymbol{m}_{t+1}) &+ p_t(\mu_{t+1}, n_t + 1 | \mu_t, n_t, a_t) v_{t+1}(\mu_{t+1}, n_t + 1, \boldsymbol{m}_{t+1}) \end{aligned} \tag{5}$$

The expansion is obtained based on our binary definition of surprise, which leads to the transition from $n_t$ being either $n_{t+1} = 0$ if surprise occurs, or $n_{t+1} = n_t + 1$, otherwise. Using the above definitions, we can now show that the monotonicity of $v_t(\mu_t, n_t, \boldsymbol{m}_t)$ and the superadditivity of $H_t(\mu_t, n_t, \boldsymbol{m}_t, a_t)$ are preserved for all decision epochs.

**Proposition 1** Suppose that Assumptions 1-4 hold. Then the following statements are true $\forall t = 0, \ldots, T - 1$:

*1.* $v_t(\mu_t, n_t, \boldsymbol{m}_t)$ *is:*

*(i) nonincreasing in* $\mu_t$ *for any* $n_t$ *and* $\boldsymbol{m}_t$

*(ii) nondecreasing in* $n_t$ *for any* $\mu_t$ *and* $\boldsymbol{m}_t$.

*2.* $\mathcal{H}_t(\mu_t, n_t, \boldsymbol{m}_t, a_t)$ *is:*

*(i) superadditive in* $\mu_t$ *and* $a_t$ *for any* $n_t$ *and* $\boldsymbol{m}_t$,

*(ii) superadditive in* $n_t$ *and* $a_t$ *for any* $\mu_t$ *and* $\boldsymbol{m}_t$.

Before proving Proposition 1-1, we need the following Lemma.

**Lemma 1** Let two reverse cumulative distribution functions be such that:

$$\sum_{j=k} p_t(j|i) \leq \sum_{j=k} p_t'(j|i) \tag{6}$$

for all $k$ .

(i) For a nonincreasing function $v(j)$, we have:

$$\sum_{j=k} p_t(j|i)v(j) \geq \sum_{j=k} p_t'(j|i)\, v(j) \tag{7}$$

(ii) The reverse inequality holds for a nondecreasing function $v(j)$.

**Proof of Lemma 1**

**Proof of Lemma 1(i).** The definition of reverse cumulative probability functions implies that the two functions are equal at $k = k_{min}$. Hence:

$$\Delta^{(1)} = \sum_{j=k_{min}} p_t'(j|i) - \sum_{j=k_{min}} p_t(j|i) = 0 \tag{8}$$

Since $v(j)$ is a nonincreasing function, we can write:

$$\Delta^{(2)} = \left\{ \sum_{j=k_{min}} p_t'(j|i) - \sum_{j=k_{min}} p_t(j|i) \right\} v(k_{min})$$

$$\geq p_t'(k_{min}|b)v(k_{min}) - p_t(k_{min}|i)v(k_{min}) + \left\{ \sum_{j=k_{min}+1} p_t'(j|i) - \sum_{j=k_{min}+1} p_t(j|i) \right\} v(k_{min}+1)$$

$$\geq \sum_{j=k_{min}}^{k_{min}+1} p_t'(j|i)v(j) - \sum_{j=k_{min}}^{k_{min}+1} p_t(j|i)v(j) + \left\{ \sum_{j=k_{min}+2} p_t'(j|i) - \sum_{j=k_{min}+2} p_t(j|i) \right\} v(k_{min}+2) \tag{9}$$

In a similar way, it follows that:

$$\Delta^{(2)} \geq \sum_{j=k_{min}} p_t'(j|i)\, v(j) - \sum_{j=k_{min}} p_t(j|i)v(j) \tag{10}$$

Hence:

$$\sum_{j=k} p_t(j|i)v(j) \geq \sum_{j=k} p_t'(j|i)v(j) \; \blacksquare \tag{11}$$

The proof of Lemma 1(ii) is similar and hence omitted.

**Proof of Proposition 1-1**

**Proof of Proposition 1-1 (i).** We prove the proposition using backward induction. Since $v_{T-1}(\mu_{T-1}, n_{T-1}, \boldsymbol{m}_{T-1}) = r_{T-1}(\mu_{T-1}, n_{T-1}, \boldsymbol{m}_{T-1})$, the result holds for $t = T-1$ by Assumption 1(i). Let's assume $v_{t+1}(\mu_{t+1}, n_{t+1}, \boldsymbol{m}_{t+1})$ is nonincreasing in $\mu_{t+1}$. By assumption, there exists an $a_t^* \in \mathcal{A}_t(\mu_t, n_t, \boldsymbol{m}_t)$ which attains the maximum in:

$$v_t(\mu_t, n_t, \boldsymbol{m}_t) = \max_{a_t \epsilon \mathcal{A}_t} \left\{ \bar{r}_t(\mu_t, n_t, \boldsymbol{m}_t, a_t) + \lambda^{\Delta t} \sum_{\mu_{t+1}} p_t(\mu_{t+1}, \cdot \mid \mu_t, n_t, a_t) v_{t+1}(\mu_{t+1}, \cdot, \boldsymbol{m}_{t+1}) \right\} \quad (12)$$

hence:

$$v_t(\mu_t, n_t, \boldsymbol{m}_t) = \bar{r}_t(\mu_t, n_t, \boldsymbol{m}_t, a_t^*) + \lambda^{\Delta t} \sum_{\mu_{t+1}} p_t(\mu_{t+1}, \cdot \mid \mu_t, n_t, a_t^*) v_{t+1}(\mu_{t+1}, \cdot, \boldsymbol{m}_{t+1}) \quad (13)$$

Let $\mu_t^- \leq {\mu_t}^+$. Since $r_t(\mu_t, n_t, \boldsymbol{m}_t, a_t^*)$ is nonincreasing in $\mu_t$, then $r_t(\mu_t^+, n_t, \boldsymbol{m_t}, a_t^*) \leq r_t(\mu_t^-, n_t, \boldsymbol{m_t}, a_t^*)$. By Assumption 1(i), Assumption 2(i), induction hypothesis, and Lemma 1(i), we have:

$$v_t(\mu_t^+, n_t, \boldsymbol{m}_t) = \bar{r}_t(\mu_t^+, n_t, \boldsymbol{m_t}, a_t^*) + \lambda^{\Delta t} \sum_j p_t(j, \cdot \mid \mu_t, n_t, a_t^*) v_{t+1}(j, \cdot, \boldsymbol{m}_{t+1})$$

$$\leq \bar{r}_t(\mu_t^-, n_t, \boldsymbol{m}_t, a_t^*) + \lambda^{\Delta t} \sum_j p_t(j, \cdot \mid \mu_t^-, n_t, a_t^*) v_{t+1}(j, \cdot, \boldsymbol{m}_{t+1})$$

$$\leq \max_{a_t \epsilon \mathcal{A}_t} \left\{ \bar{r}_t(\mu_t^-, n_t, \boldsymbol{m}_t, a_t) + \lambda^{\Delta t} \sum_j p_t(j, \cdot \mid \mu_t^-, n_t, a_t) v_{t+1}(j, \cdot, \boldsymbol{m}_{t+1}) \right\} = v(\mu_t^-, n_t, \boldsymbol{m}_t) \quad (14)$$

Hence $v_t(\mu_t, n_t, m_t)$ is nonincreasing in $\mu_t$.■

The proof of Proposition 1-1(ii) is similar (using Assumptions 1(ii) and 2(ii) as well as Lemma 1(ii)) and hence omitted.

**Proof of Proposition 1-2**

**Proof of Proposition 1-2 (i).** By Assumption 4(i) and the definition of subadditivity, for $a_t^- \leq a_t^+$ and $\mu_t^- \leq \mu_t^+$ we have:

$$\sum_{\mu_{t+1}=k} \{p_t(\mu_{t+1},\cdot\,|\mu_t^+, n_t, a_t^+) + p_t(\mu_{t+1},\cdot\,|\mu_t^-, n_t, a_t^-)\}$$
$$\leq \sum_{\mu_{t+1}=k} \{p_t(\mu_{t+1},\cdot\,|\mu_t^+, n_t, a_t^-) + p_t(\mu_{t+1},\cdot\,|\mu_t^-, n_t, a_t^+)\} \quad (15)$$

In our case $a_t^+ = i$, which means prescribing medication $i$ and $a_t^- = 0$, which means doing nothing.

By Proposition 1-1(i), $v_{t+1}(\mu_{t+1}, n_{t+1}, m_{t+1})$ is nonincreasing in $\mu_{t+1}$. Hence applying Lemma 1(i) yields:

$$\sum_{\mu_{t+1}=\underline{B}} \{p_t(\mu_{t+1},\cdot\,|\mu_t^+, n_t, i) + p_t(\mu_{t+1},\cdot\,|\mu_t^-, n_t, 0)\}\, v_{t+1}(\mu_{t+1},\cdot, \boldsymbol{m}_{t+1}) \geq$$
$$\sum_{\mu_{t+1}=\underline{B}} \{p_t(\mu_{t+1},\cdot\,|\mu_t^+, n_t, 0) + p_t(\mu_{t+1},\cdot\,|\mu_t^-, n_t, i)\} v_{t+1}(\mu_{t+1},\cdot, \boldsymbol{m}_{t+1}) \quad (16)$$

Therefore, $\sum_{\mu_{t+1}} p_t(\mu_{t+1},\cdot\,|\mu_t, n_t, a_t) v_{t+1}(\mu_{t+1},\cdot, \boldsymbol{m}_{t+1})$ is superadditive in $\mu_t$ and $a_t$ for each $t$ including $t+1$. From Assumption 3(i) , $r_t(\mu_t, n_t, \boldsymbol{m}_t, a_t)$ is superadditive in $\mu_t$ and $a_t$, and since the sum of superadditive functions is superadditive, then $H_t(\mu_t, n_t, \boldsymbol{m}_t, a_t)$ is superadditive in $\mu_t$ and $a_t$. ■

The proof of Proposition 2-1(ii) is similar (using Assumptions 4(ii) and 3(ii), Proposition 1-1(ii), and Lemma 1(ii)) and hence omitted.

Given the preservation of the monotonicity and superadditivity properties, we can now prove the monotonicity of optimal policy $a_t^*(\mu_t, n_t, \boldsymbol{m}_t)$.

To prove Theorem 1, we need Lemma 2.

**Lemma 2** Suppose $\mathcal{H}_t(\mu_t, n_t, \boldsymbol{m}_t, a_t)$ is a superadditive function in $\mu_t$ and $a_t$, and $\max_{a_t \epsilon \mathcal{A}_t} [\mathcal{H}_t(\mu_t, n_t, \boldsymbol{m}_t, a_t)]$ exists. Then $a_{1t}^*(\mu_t, n_t, \boldsymbol{m}_t) = \max\{a_t' \in argmax\, [\mathcal{H}_t(\mu_t, n_t, \boldsymbol{m}_t, a_t)]\}$ is monotone nondecreasing in $\mu_t$.

**Proof of Lemma 2** The proof is the same as the proof of Lemma 4.7.1 in Puterman (2005) and hence omitted. ■

**Proof of Theorem 1-1** (i.e., monotonicity of optimal policy $a_t^*(\mu_t, n_t, m_t)$ with respect to perceived health state $\mu_t$**).** Since we have proved the superadditivity of $\mathcal{H}_t(\mu_t, n_t, \boldsymbol{m_t}, a_t)$ in Proposition 1-2(i), then the proof of Theorem 1-1 follows from Lemma 2. ■

The proof of Theorem 1-2 (i.e., monotonicity of optimal policy $a_t^*(\mu_t, n_t, m_t)$ with respect to the belief strength $n_t$) is similar (using Proposition 1-2(ii) and Lemma 2) and hence omitted.

**Theorem 2.** Suppose that Assumptions 1-4 hold $\forall t = 0, \ldots, T-1$. Then $\mu_t^*(n_t)$ is

1. nonincreasing with $n_t$.
2. nonincreasing with $t$.

**Proofs**. First, let us define:

$$\Delta\mathcal{H}_t(\mu_t, n_t, \boldsymbol{m}_t, a_t = i) = \mathcal{H}_t(\mu_t, n_t, \boldsymbol{m}_t, i) - \mathcal{H}_t(\mu_t, n_t, \boldsymbol{m}_t, 0) \tag{17}$$

$\Delta H_t$ quantifies the benefit of more aggressive treatment (i.e., prescribing medication $i$) versus less aggressive treatment (i.e., doing nothing) for the same $\mu_t, n_t$ and $\boldsymbol{m}_t$. Therefore the first statement requires $\Delta\mathcal{H}_t(\mu_t, n_t^-, \boldsymbol{m}_t, a_t) \leq \Delta\mathcal{H}_t(\mu_t, n_t^+, \boldsymbol{m}_t, a_t)$, which means that for the same $\mu_t$ and $\boldsymbol{m}_t$ the expected benefits of prescribing medication increases with $n_t$ (i.e., the physician's confidence in $\mu_t$). Therefore:

$$\mathcal{H}_t(\mu_t, n_t^-, \boldsymbol{m}_t, i) - \mathcal{H}_t(\mu_t, n_t^-, \boldsymbol{m}_t, 0) \leq \mathcal{H}_t(\mu_t, n_t^+, \boldsymbol{m}_t, i) - \mathcal{H}_t(\mu_t, n_t^+, \boldsymbol{m}_t, 0) \tag{18}$$

Rearranging the inequality, we will have:

$$\mathcal{H}_t(\mu_t, n_t^+, \boldsymbol{m}_t, i) + \mathcal{H}_t(\mu_t, n_t^-, \boldsymbol{m}_t, 0) \geq \mathcal{H}_t(\mu_t, n_t^-, \boldsymbol{m}_t, i) + \mathcal{H}_t(\mu_t, n_t^+, \boldsymbol{m}_t, 0) \tag{19}$$

This is the definition of superadditivity of $H_t$ with $n_t$ and $a_t$ which was proved in Proposition 1-2(ii). ■

The proof of Theorem 2-2 is similar and hence omitted. In summary, Theorem 2-2 requires that $\Delta\mathcal{H}_t(\mu_t, n_t, \boldsymbol{m}_t, a_t) \leq \Delta\mathcal{H}_{t+1}(\mu_t, n_t, \boldsymbol{m}_t, a_t)$, which means that for the same $\mu_t, n_t$ and $\boldsymbol{m}_t$ the expected benefits of prescribing medication increases with $t$. This makes sense given that 1) age is a risk factor for CVD and 2) with aging, there will be lower remaining negative impacts of medication side-effects. This is also consistent with the clinical results in the seminal SPRINT (Systolic Blood Pressure Intervention Trial) study (SPRINT Research Group 2015) and a large empirical study, based on the observations of 8.8 million patients in the United States, which recommend lowering BP thresholds with aging (Fletcher et al. 2017).

## 3. Violations of Assumptions

In this section, we quantify the violations of the assumptions we made to characterize the structure of optimal policies for **MDP$^{\mathbf{SIL}}$**. To do so, we introduce notation $\varsigma^{Assumption\#}$ to define and evaluate violations of the corresponding assumption.

First, we discuss the interpretations of the IFRness property of $P_t$.

a. The IFRness of $P_t$ in $\mu_t$ implies that the reverse cumulative function $\varrho_1(\mu_t, n_t)$ is nondecreasing in $\mu_t$ for all $n_t, \mu'$ where

$$\varrho_1(\mu_t, n_t) = \sum_{\mu_{t+1} \geq \mu'} p_t(\mu_{t+1}, \cdot \, | \mu_t, n_t, a_t) \tag{20}$$

and

$$\begin{aligned} p_t(\mu_{t+1}, \cdot \, | \mu_t, n_t, a_t) &= \sum\nolimits_{n_{t+1}=0}^{n_t+1} p_t(\mu_{t+1}, n_{t+1} | \mu_t, n_t, a_t) \\ &= p_t(\mu_{t+1}, 0 | \mu_t, n_t, a_t) + p_t(\mu_{t+1}, n_t + 1 | \mu_t, n_t, a_t) \end{aligned} \tag{21}$$

b. The IFRness of $P_t$ in $n_t$ implies that the reverse cumulative function $\varrho_2(\mu_t, n_t)$ is nondecreasing in $n_t$ for all $\mu_t, n_t'$, where

$$\varrho_2(\mu_t, n_t) = \sum_{n_{t+1} \geq n'} p_t(\cdot, n_{t+1} | \mu_t, n_t, a_t) \tag{22}$$

and

$$p_t(\cdot, n_{t+1} | \mu_t, n_t, a_t) = \sum_{\mu_{t+1}} p_t(\mu_{t+1}, n_{t+1} | \mu_t, n_t, a_t) \tag{23}$$

In our study, the first requirement (i.e., the IFRness with $\mu_t$) means that the higher the level of BP is, the more likely it is transitioning to higher BP levels or terminal events. The procedure we have followed for the normal probability discretization (introduced by Roy (2003)) guarantees the preservations of the IFR property of the normal distribution. Below, we have also numerically assessed the maximum violation of the IFR property.

For the second requirement (i.e., the IFRness with $n_t$), first, recall that:

$$n_t' = \begin{cases} 0, & \text{if facing surprise} \\ n_t + 1, & \text{otherwise} \end{cases} \tag{24}$$

In the second case (i.e., $k_{2t} = n_t' = n_t + 1$), $\varrho_2(\mu_t, n_t)$ can be interpreted as the probability of facing no-surprise. Therefore, the second requirement means that the higher the belief strength (which also indicates a larger number of visits since the last surprise), the less likely it will be facing a surprise, which makes sense for two reasons. First, higher $n_t$ means increased BP stabilization (around a certain level), which leads to a lower likelihood of facing a surprise. Second, higher $n_t$ implies narrower normal distribution with which the BP observations will be generated. This, in turn, leads to the lower likelihood of facing a surprise, which is defined as the deviation from the mean of the belief distribution. The requirement holds in the first case (i.e., $n_t' = 0$), too, as in this case, $\varrho_2(\mu_t, n_t)$ is equal to 1 for all $n_t$.

Now, we define the violations as follows:

$$\varsigma^{2(i)} = \max_{\mu', \mu_t, n_t, a_t, t} \left\{ \max\left\{ 0, \sum_{\mu_{t+1} \geq \mu'} p_t(\mu_{t+1}, \cdot \mid \mu_t, n_t, a_t) - \sum_{\mu_{t+1} \geq \mu'} p_t(\mu_{t+1}, \cdot \mid \mu_t + \Delta B^{SIL}, n_t, a_t) \right\} \right\} \tag{25}$$

$$\varsigma^{2(ii)} = \max_{n', \mu_t, n_t, a_t, t} \left\{ \max\left\{ 0, \sum_{n_{t+1} \geq n'} p_t(\cdot, n_{t+1} \mid \mu_t, n_t, a_t) - \sum_{n_{t+1} \geq n'} p_t(\cdot, n_{t+1} \mid \mu_t, n_t + 1, a_t) \right\} \right\} \tag{26}$$

To define $\varsigma^{4(i)}$ and $\varsigma^{4(ii)}$, define:

$$\Delta q_{1t}(\mu_t) = q_{1t}(\mu', \mu_t, n_t, i) - q_{1t}(\mu', \mu_t, n_t, 0) \tag{27}$$

$$\Delta\Delta q_{1t} = \Delta q_{1t}(\mu_t + 1) - \Delta q_{1t}(\mu_t) \tag{28}$$

and

$$\Delta q_{2t}(n_t) = q_{2t}(n', \mu_t, n_t, i) - q_{2t}(n', \mu_t, n_t, 0) \tag{29}$$

$$\Delta\Delta q_{2t} = \Delta q_{2t}(n_t + 1) - \Delta q_{2t}(n_t) \tag{30}$$

Therefore:

$$\varsigma^{4(i)} = \max_{\mu_t, t}\{\max\{0, \Delta q_{1t}(\mu_t + 1) - \Delta q_{1t}(\mu_t)\}\}$$

$$= \max_{\mu', \mu_t, n_t, i, t}\Big\{\max\Big\{0, \{q_{1t}(\mu', \mu_t + \Delta B^{SIL}, n_t, i) - q_{1t}(\mu', \mu_t + \Delta B^{SIL}, n_t, 0)\}$$

$$- \{q_{1t}(\mu', \mu_t, n_t, i) - q_{1t}(\mu', \mu_t, n_t, 0)\}\Big\}\Big\}$$

$$= \max_{\mu', \mu_t, n_t, i, t}\left\{\max\left\{0, \left[\sum_{\mu_{t+1} \geq \mu'} p_t(\mu_{t+1}, \cdot\, |\mu_t + \Delta B^{SIL}, n_t, i) - \sum_{\mu_{t+1} \geq \mu'} p_t(\mu_{t+1}, \cdot\, |\mu_t + \Delta B^{SIL}, n_t, 0)\right]\right.\right.$$

$$\left.\left. - \left[\sum_{\mu_{t+1} \geq \mu'} p_t(\mu_{t+1}, \cdot\, |\mu_t, n_t, i) - \sum_{\mu_{t+1} \geq \mu'} p_t(\mu_{t+1}, \cdot\, |\mu_t, n_t, 0)\right]\right\}\right\} \tag{31}$$

and

$$\varsigma^{4(ii)} = \max_{n_t, t}\{\max\{0, \Delta q_{2t}(n_t) - \Delta q_{2t}(n_t + 1)\}\}$$

$$= \max_{n', \mu_t, n_t, i, t}\{\max\{0, [q_{2t}(n', \mu_t, n_t, i) - q_{2t}(n', \mu_t, n_t, 0)]$$

$$- [q_{2t}(n', \mu_t, n_t + 1, i) - q_{2t}(n', \mu_t, n_t + 1, 0)]\}\}$$

$$= \max_{n', \mu_t, n_t, i, t}\left\{\max\left\{0, \left[\sum_{n_{t+1} \geq n'} p_t(\cdot, n_{t+1}|\mu_t, n_t, i) - \sum_{n_{t+1} \geq n'} p_t(\cdot, n_{t+1}|\mu_t, n_t, 0)\right]\right.\right.$$

$$\left.\left. - \left[\sum_{n_{t+1} \geq n'} p_t(\cdot, n_{t+1}|\mu_t, n_t + 1, i) - \sum_{n_{t+1} \geq n'} p_t(\cdot, n_{t+1}|\mu_t, n_t + 1, 0)\right]\right\}\right\} \tag{32}$$

Finally, the violations of Assumption 3 can be calculated as follows:

$$\varsigma^{3(i)} = \max_{\mu_t,i,t}\{\max\{0, [\bar{r}_t(\mu_t + 1, n_t, \boldsymbol{m}_t, i) + \bar{r}_t(\mu_t, n_t, \boldsymbol{m}_t, 0)] - [\bar{r}_t(\mu_t + 1, n_t, \boldsymbol{m}_t, 0) + \bar{r}_t(\mu_t, n_t, \boldsymbol{m}_t, i)]\}\} \quad (33)$$

$$\varsigma^{3(ii)} = \max_{n_t,i,t}\{\max\{0, [\bar{r}_t(\mu_t, n_t + 1, \boldsymbol{m}_t, i) + \bar{r}_t(\mu_t, n_t, \boldsymbol{m}_t, 0)] - [\bar{r}_t(\mu_t, n_t + 1, \boldsymbol{m}_t, 0) + \bar{r}_t(\mu_t, n_t, \boldsymbol{m}_t, i)]\}\} \quad (34)$$

Table C-1 summarizes the maximum violations of the assumptions we used in this study.

**Table C-1.** Violations of assumptions

| **Assumption#** | **Equation #** | **Maximum Violation ($\varsigma^{Assumption\#}$)** |
|---|---|---|
| **2(i)** | 25 | 0.003 |
| **2(ii)** | 26 | 0.000 |
| **3(i)** | 33 | 0.091 |
| **3(ii)** | 34 | 0.073 |
| **4(i)** | 31 | 0.054 |
| **4(ii)** | 32 | 0.000 |

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